Project Report

FILOX & FILKGC: LLM-Driven Ontology Expansion and Knowledge Graph Construction for Indian Legal Texts

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in

Computer Science

Submitted by

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Declaration

This is to certify that this project report titled **"FILOX & FILKGC: LLM-Driven Ontology Expansion and Knowledge Graph Construction for Indian Legal Texts"** is being submitted to the Department of Computer Science, University of Delhi, New Delhi in partial fulfillment of the requirement for the award of the degree of M.Sc. (Computer Science). It is a record of bonafide work carried out by us under the supervision of Prof. Vasudha Bhatnagar and Dr. Vikas Kumar. The matter embodied in the declaration has not been submitted in part or full to any university or institution for the award of any degree or diploma.

Date: June 29, 2025 **Harsh Yadav (23234747026)**

Place: New Delhi

Certificate

This is to certify that this project report titled **"FILOX & FILKGC: LLM-Driven Ontology Expansion and Knowledge Graph Construction for Indian Legal Texts"** submitted to the Department of Computer Science, University of Delhi, in partial fulfillment of the requirement for the award of the degree of M.Sc. (Computer Science), is a research work carried out by Harsh Yadav (23234747026) under the joint supervision of Prof. Vasudha Bhatnagar and Dr. Vikas Kumar. This work has not been submitted in part or full to any other University or Institution for the award of any degree or diploma, to the best of our knowledge.

Prof. Neelima Gupta	**Prof. Vasudha Bhatnagar**	**Dr. Vikas Kumar**
(Head)	(Supervisor)	(Supervisor)

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Chapter 1

Introduction

As of 29 June 2025, over 56.57 crore (565.65 million) orders and judgments have been recorded from India's computerized courts, according to the National Judicial Data Grid (NJDG) [1]. The NJDG collects data from 733 districts and 3,612 total court complexes. These numbers show the vast amount of legal data that is being generated from a very long period. This is a great resource for the community of current researchers and legal practitioners, but the issue lies with the unstructured nature of these documents. It is very complex and difficult to navigate through these to find the information about important events, judges, appellant, defendant, precedents, statutes, provisions, the final decision of the judge or bench, etc. Due to this nature, legal practitioners have no options but to dive deeper into each and every judgment to know whether it is similar to the case they are defending or not and end up wasting a lot of their crucial time. This task is often referred to as case recommendation in legal terminology and is almost used every day by a practitioner or a researcher. More tasks like this include question-answering by creating a legal chatbot, document summarization, semantic search, fine-tuning Large Language Models (LLMs), decision support, risk assessment, and many more.

But, to perform any of the tasks mentioned above, the data must be in a structured format. For decades, researchers from all over the world have been working to develop state-of-the-art methods and frameworks to structure such types of complex and unstructured text data, and the field of research has been known as Information Extraction (IE), under the umbrella of the broader research area of Natural Language Processing (NLP). IE task can be further segregated into 3 main research areas: Named Entity Recognition (NER), Relation Extraction (RE), and Event Extraction (EE). We will discuss these in great detail in further sections of the report.

However, the key point here is that these areas have been researched for a long time for the general domain text and hence have limitations when performed in domain-

[1] `https://njdg.ecourts.gov.in/njdg_v3//`

specific settings. Another concern is that most of the work has been done on traditional methods, such as rule-based and classical Machine Learning (ML) techniques, which require huge amounts of annotated data to be trained on, another limitation from which the legal domain is already suffering.

However, recently, with the advancements in the field of Artificial Intelligence (AI), many giant companies have introduced their large language models, which are pre-trained on a massive amount of text corpora and have context of almost every discipline that exists. Some of these LLMs are proprietary, i.e., the model weights, codes, and architecture are kept hidden from the public, and to take their service, one has to send the query to their server. A few examples include OpenAI's ChatGPT models, Anthropic's Claude models, Google's Gemini models, etc. While some of the companies are also open-source in this race, the models are available to download and run locally, free of cost. The examples include Meta's Llama models, Mistral's models, DeepSeek's R1 models, etc. The legal IE community urgently needs such systems or frameworks that can utilize these LLMs suitably to perform IE tasks on legal documents to improve the performance of the legal IE field without the requirement of domain-specific annotated data.

As Isaac Newton wrote in a letter to Robert Hooke in 1675, *"If I have seen further than others, it is by standing upon the shoulders of giants"*. Inspired by this perspective, our work builds upon prior advancements by integrating the **DeepSeek R1** model into our proposed approach. Among all the possible options, we have used the open-source DeepSeek R1 model because of its state-of-the-art reasoning capabilities. Its release in January 2025 took the internet by storm. Its scores in natural language, math, and reasoning were close to OpenAI's proprietary o1 model and outperformed in some coding and mathematics tests.

But the power of these LLMs of having wide knowledge of disciplines can often become their limitations too when used in domain-specific settings. For a given legal document, a typical LLM can often extract the entities or can label the extracted entities with *entity-types*, irrelevant to the legal domain or terminology. To handle this, we need to provide some ontology that can guide the model during the extraction process of entities and *entity-types*. Existing Indian legal ontologies are very small, having very few *entity-types* and relations, which are not sufficient to extract all legal terms in a document. Hence, we have proposed a framework that can expand the existing NyOn[1] ontology based on a data-driven approach. We have shown the expansion on 114 Supreme Court judgments using the proposed framework. Additionally, we have developed another framework, which can perform IE using any such ontology. Frameworks of both the tasks utilize the power of LLMs to accomplish their respective goals.

Please note that the proposed frameworks are not dependent on a specific model and can easily be used with the model's current or future alternatives.

1.1 Information Extraction (IE)

Information Extraction (IE) is the process of extracting information relevant to some interest, which includes extracting named entities, relations, and events [2]. IE is an important research area of Natural Language Processing (NLP), and the goal of IE systems is to automatically extract structured information from unstructured and semi-structured machine-readable documents [3].

Triplet Extraction:

These IE systems extract structured information in the form of triplets. Each triplet includes two entities, a subject and an object, and a relation linking these entities. This triplet can then be used to construct a knowledge graph [4]. One example could be ['Faculty of Mathematical Sciences', 'FacultyOf', 'University of Delhi']; here, 'Faculty of Mathematical Sciences' is the subject entity, which is associated with the object entity 'University of Delhi' in the relation as 'FacultyOf'.

As mentioned in the introduction, IE involves research in three most discussed areas or sub-tasks: (i) named entity recognition, (ii) relationship extraction, and (iii) event extraction. **Named Entity Recognition (NER)** includes two tasks: entity identification, which involves identifying head and tail entities, and entity typing, which focuses on assigning types to these identified entities. The second research area, **Relationship Extraction (RE),** focuses on finding relationships between different entities, which often act as edges of a knowledge graph. The third research area, **Event Extraction (EE),** focuses on identifying and classifying the triggered word and type that most clearly represents the occurrence of an event [5].

NER and RE are considered the two minimum requirements in the pipeline to create a triplet. Whereas some works also refer to this as the Relation Triplet (RT) task, which comes under the RE task, independent of the NER task. Optionally, triplet canonicalization is a subsequent phase in most IE pipelines, aimed at reducing ambiguity and redundancy in the extracted triplets that may have been classified as distinct due to variations in the synonyms or terms used for the relation.

1.2 Knowledge Graphs (KGs)

Google was the first one to introduce the term *knowledge graph* in 2012 to build a model of the world [6]. In our application, knowledge graphs are used as a database to store the extracted information in the form of nodes and edges for representing concepts and relationships, respectively, in a structured way, as shown in Figure 1.1.

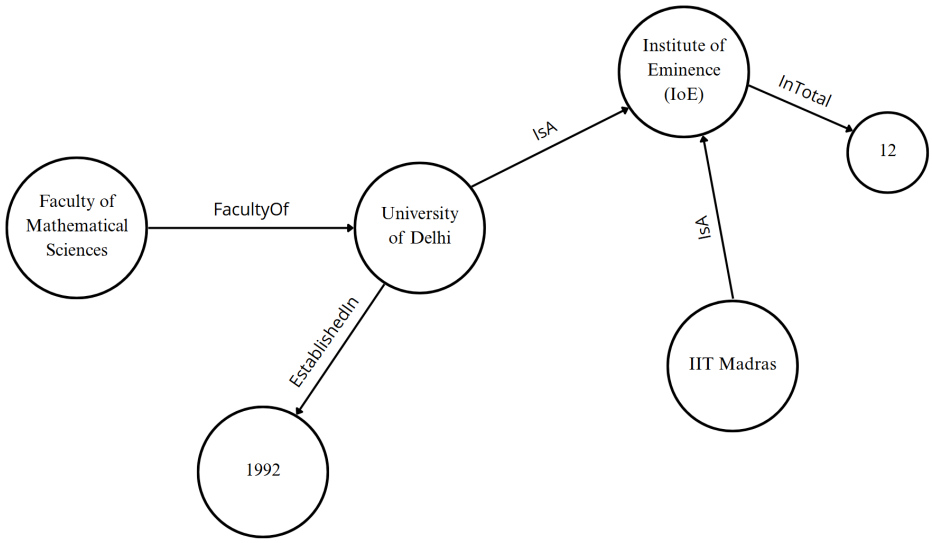

Figure 1.1: An example of entities and relations in a KG

Relational Databases vs Graph Databases

One valid question that arises here is, if we have to store the extracted information somewhere, why not use relational databases such as MySQL? Why do we necessarily need to store it as graphs? The answer to it is described in the paragraph below:

Knowledge graphs offer distinct advantages over relational databases in managing legal data. Graphs are able to keep all the information about an entity in a single node and show related information by different nodes connected to it [7]. They naturally capture complex relationships and hierarchies between entities, such as legal actions, relationships between individuals, etc., which are challenging to represent with relational tables. They try to overcome the limitations imposed by traditional DB models with respect to capturing the inherent graph structure of data, where the interconnectivity of data is an important aspect [8]. Legal documents contain complex, interconnected information, and if stored in relational databases, it will require complex joins to represent such relationships because relational DBs work on flat, predefined tables. In contrast to which, KGs are flexible and can store these relationships easily. Furthermore, KGs can be adapted to dynamic schemas based on the demand of the constantly evolving legal domain. This flexibility allows the addition of new entities and relationships without worrying about the underlying table schema.

In addition, various graph algorithms can be applied to these KGs to analyze the legal document in detail. Not only on just one document, graph algorithms can also be used to compute similarity between two cases by constructing KGs of both the cases' documents. KGs also allow more context-aware querying, which helps in precise searching within the document in comparison to keyword-based searching. Relational DBs, on the other hand, lack inherent context awareness and thus require complex SQL queries to handle such relationships, which can be inefficient.

Broad Classification of Knowledge Graphs

Knowledge graphs can be divided on the basis of a lot of factors, like their implementation (Resource Description Framework (RDF)) or construction technique (rule-based knowledge graphs and learning-based knowledge graphs), but for our purposes, we are dividing them on the basis of domain scope: (i) general knowledge graphs and (ii) domain-specific knowledge graphs [9]:

General KGs: The aim of these graphs is to organize knowledge of various fields, independent of a single domain.

Some well-known examples of general KGs include: **Google Knowledge Graph** [2] which helps Google Search in understanding the context of search queries and providing precise information as the response. **DBpedia** [10] which uses Wikipedia's vast text corpus to create a structured KG, so that researchers can perform large-scale semantic analysis on the same. **YAGO (Yet Another Great Ontology)** [11] which combines structured data from text corpuses of Wikipedia and WordNet to help in linguistic and cognitive sciences research.

Domain-Specific KGs: Unlike general KGs, these graphs focus on knowledge within a specialized field, such as biology, law, or finance [12]. The construction of these graphs is designed to capture the intricacies and depth of the specific domain. Due to the lack of a 'one-size-fits-all' schema or ontology that can be applied to address real-life problems, efforts are continuing to be made to establish, polish, and improve domain-specific KGs in several domains of knowledge. For example, [13] presented K12EduKG, a domain-specific KG based on K-12 educational subjects. Domain-specific educational data (Chinese curriculum standards of mathematics) was the source of knowledge that was used in K12EduKG. [13] also pointed out the limitations of general KGs, i.e., they are good for real-world entities like PERSON, ORGANIZATION, and LOCATION, but they face difficulties when applied to the educational domain. This is because educational concepts are often abstract and cannot be easily defined or extracted, unlike the concrete entities in generic graphs. Furthermore, the relationships between educational concepts are more cognitive and implicit, which makes them difficult to derive from the literal meanings of the text.

Similar to the above example, many other examples of different domains are listed below: **GeoNames** [3] which provides geographical data for applications like mapping and geographic information systems (GIS). **MusicBrainz** [4] which uses global music metadata to develop recommendation engines so that it can be used by music streaming platforms. **CIA World Factbook** [5] which organizes geopolitical data so that it can be used for academic

[2] https://en.wikipedia.org/wiki/Google_Knowledge_Graph
[3] https://en.wikipedia.org/wiki/GeoNames
[4] https://en.wikipedia.org/wiki/MusicBrainz
[5] https://en.wikipedia.org/wiki/The_World_Factbook

research and government policymaking. **DrugBank** [6] which structures detailed drug and pharmaceutical information to support medical research and healthcare applications. **Legal KGs** which help in legal research and similar case recommendations [14].

In this report we will only be focusing on the legal domain knowledge graph construction.

1.3 Legal Knowledge Graphs (LKGs)

The legal professionals, such as lawyers, judges, researchers, etc., are all dependent on vast amounts of data. They use this data to make informed decisions, to make arguments on its basis, and to study the cases. In the era of technology-driven innovation, LKGs are becoming powerful tools that promise to revolutionize the practice of law.

Some of the potential applications of knowledge graphs in legal domain include [15], [16], [14], [17], [18], [19]: **Legal Judgment Prediction**, **Similar Case Matching**, **Legal Question-Answering**, **Legal Document Summarization** and **Finetuning LLMs**.

Challenges in Legal Knowledge Graph Construction

While knowledge graphs are highly promising for the legal domain, there are various factors that make their creation challenging: Firstly, the **complexity of the legal language**: the judgments are written in a very complicated manner, which makes them ambiguous. Also, much jargon is used in the text, which makes it even harder to extract information. Secondly, laws keep changing with the time, so the KGs need timely **maintenance** and updating. Thirdly, there are very **few high-quality triplet datasets** present to construct the KGs. The existing ones are extracted from those ontologies, which are very small and do not cover various legal relationships and entities. Lastly, majority of the current ontologies are based on **super-class and sub-class structures**, which makes the extraction process complex and does not add much value to the KGs.

If we can somehow fix these issues and overcome these challenges, we can use the KGs to revolutionize the legal domain in the near future. With the emerging AI-driven reasoning, eventually constructing legal KGs will become more accurate and usable, which will help in efficient legal research. Researchers and practitioners need to continue to collaborate on overcoming these obstacles by developing new methods for legal data extraction, methods for expanding ontologies, and creating more robust datasets to help the construction of sophisticated, domain-specific legal KGs.

[6] `https://en.wikipedia.org/wiki/DrugBank`

1.4 Objective

With the exponential growth of unstructured data, integrating generative AI and LLMs with traditional approaches has become essential. Millions of legal articles, cases, and judgments are produced by courts and sub-courts in India. This complex, unstructured data is invaluable for legal practitioners yet often challenging to navigate effectively.

There are two objectives of our research. Firstly, to create a framework that can expand the existing legal ontology by using LLMs in a data-driven manner. Secondly, to develop an LLM-based framework that can utilize any such predefined ontology to extract information from legal documents to construct legal KGs. These KGs then can directly be analyzed or can be used to aid downstream tasks such as legal case summarization, question-answering, etc.

Our contributions are as follows:

- **FILOX** (**F**ramework for **I**ndian **L**egal **O**ntology E**x**pansion): Designed a framework leveraging Large Language Models (LLMs) to expand Indian legal ontology.
- Extended the existing Indian legal ontology NyOn [1] to **87 *entity-types*** and **320 relations**, extracted from **114 Indian judgments** using the FILOX.
- Built another framework, **FILKGC** (**F**ramework for **I**ndian **L**egal **K**nowledge **G**raph **C**onstruction): Designed to perform LLM-based Information Extraction (IE) on legal documents by utilizing the expanded ontology, to construct legal Knowledge Graphs (KGs).
- Extracted **36,070 triplets** from 114 judgments to construct rich KGs using FILKGC. Entity and relation canonicalization could not be performed because of time and compute constraints.

To the best of our knowledge, this is the first research work in the Indian legal domain which utilizes LLMs for ontology expansion and KGC tasks. No work has been done till now to expand the Indian legal ontologies using the LLM automated frameworks (FILOX and FILKGC) we have proposed, which makes it novel in the Indian legal field. Our work paves the way for building efficient and insightful LLM automated ontology expansion system and KGC system, aiding legal practitioners and researchers in unlocking the value hidden within complex legal texts, specially those on which traditional systems performed poor because of insufficient annotated data for training the models.

1.5 Organization of the Report

The rest of the report is organized as follows: In **Chapter 2** we provide an in-depth analysis of relevant research papers that have informed and influenced the current project. In **Chapter 3**, we present the methodology of our proposed approaches. In **Chapter 4**, we present the implementation details and the experimental results. In **Chapter 5** we put forward our conclusions and possibilities of future work related to the project.

Appendix A contains the GitHub repository's structure, which contains scripts and outputs. It also contains the extended *entity-types* and their one-liner definitions.
Appendix C contains the extended relations and their one-liner definitions.

Chapter 2

Literature Review

In this chapter we present a literature review of information extraction (IE) systems, ontologies, and knowledge graphs. Starting from simple schema-based IE systems all the way to LLM-based generative IE systems, which are used to construct KGs.

2.1 History of IE

The research in the field of IE has emerged from the 1960s and 1970s in AI and natural language work [20]. DAPRA Message Understanding Conferences (MUC, mid-1990s) were the first large-scale recognitions that defined core IE tasks (e.g., named entity recognition of persons/organizations, template filling for events). In these conferences, DAPRA-the US defence agency organized competitions and provided funds to the research groups to pursue IE. During this era, **rule-based** and **pattern-matching systems** started to emerge. For example, FASTUS [21](a series of connected small state machines) achieved high precision by successively tokenizing text, recognizing names, and matching document-specific patterns [20]. Such systems were good at precision, but they relied heavily on expert-crafted rules or domain grammars, which meant they could miss many unexpected expressions (low recall) and required laborious manual development [22].

In the 2000s, IE shifted towards **statistical machine learning**. Models like Hidden Markov, Maximum Entropy, and Conditional Random Fields (CRF) began to replace hand-coded rules. Pioneering NER work (e.g., [23]) and the Stanford NER [24] used supervised sequence models to tag person/organization/location in text. For relation extraction, [25] made such a paradigm that does not require labeled corpora, an alternative to the works based on supervised learning.

In the 2010s, neural network based approaches started to emerge, revolutionizing the IE research. Deep learning methods (CNNs, BiLSTMs, etc.) learned features automatically,

improving performance to a greater extent [26]. Studies showed that deep models changed profoundly the rules of the game in IE, with performance rising to "unprecedented levels" over traditional ML [26]. For instance, BiLSTM-CRF architectures became standard for NER, and convolutional or LSTM models for relation and event extraction. Pretrained contextual representations (e.g., ELMo, BERT [27]) further advanced the IE field. BERT and its successors, fine-tuned on IE tasks, achieved new best results with minimal task-specific engineering [28]. The **strengths** of neural IE included high accuracy and no need for handwritten rules. The **weaknesses,** however, included the need for a huge amount of annotated data and the unexplainability of the results (harder to interpret).

More recently, Large Language Models (LLMs) like GPT-3/4 and LLAMA have begun to be applied to IE via prompting or fine-tuning. Studies like [5] discuss common backbones for generative IE. These include BART, T5, GLM, LLAMA, and GPT versions. Another work, [29] performs open IE following the 3-phase structure, Extract-Define-Canonicalize (EDC), by utilizing Mistral and GPT models. The **strengths** of LLM-based IE systems are their ability to perform IE without fine-tuning the models and thus do not require annotated datasets. Techniques like In-Context Learning (ICL), persona-based prompting, and few-shot learning further enhance their performance. The **weaknesses** include their degraded performance as compared to fine-tuned BERT and other SOTA models [5], [30].

IE techniques have not been restricted to any single domain; rather, they are applied in various domains, including medical, legal, finance, news, and intelligence. The medical domain rapidly used IE for tasks like extracting entities (diseases, drugs, genes) and relations (protein-protein, drug-drug interactions). Due to this, various biomedical NER datasets and relation extraction datasets have been released, including BC2GM, NCBI disease [31], and BioCreative CDR [32]. Tools like MetaMap [33] and cTakes [34] use medical ontologies (UMLS [35], SNOMED [36]) for dictionary-based entity recognition. Similarly, in the legal domain, legal text analysis is an active area of IE research. We will discuss Legal IE in more detail in the next section.

2.2 Legal IE

Early legal IE systems included rule-based systems, which used handcrafted patterns to extract citations, statutes, parties, and court decisions. For example, [37] developed a NER system for US legal transcripts that identified 5 *entity-types*: judges, attorneys, companies, jurisdictions, and courts.

As ML progressed, various statistical and ML methods have been proposed, including [38], and later neural methods have been applied to legal IE. Recent work applies deep learning and transformers to legal text: for instance, for legal NER and relation extraction,

various researchers have used fine-tuned BERT models to improve accuracy, such as [39], [40] for NER on German legal texts, [41] for NER on Chinese legal texts, and [42] for NER on Indian legal texts. Common limitations in legal IE have been the lack of large annotated corpora and the intricate language of law (long sentences, cross-references).

2.3 Indian Legal IE

Within the legal domain, there has been growing work specifically on Indian judiciary texts. India's legal publications (Supreme Court and High Court judgments) present an important corpus. Recent efforts include building annotated corpora and models for legal NER, rhetorical role labeling, and relation extraction in Indian case law.

One such major contribution is a large annotated corpus for legal NER developed by [42]. They created a dataset of 14,444 annotated Indian court judgment sentences and 2,126 judgment preambles with 14 legal *entity-types* (e.g., COURT, JUDGE, CASE, PARTIES, ACT, etc.). They also created a baseline transformer-based NER model to perform NER. Indian Legal Text Understanding and Reasoning (IL-TUR) [30] is another great resource that includes various IE sub-tasks. For NER specifically, they have used 105 cases and annotated them with 12 legal *entity-types*. They have also experimented with 5 different versions of the BERT model to carry out the NER task. Another work [43], constructed a KG by using 16 legal *entity-types*, 13 legal relations, and a rule-based pipeline. Another work, Nyaya Ontology (NyOn) [1], was developed to aid the construction legal KGs. It is a hierarchically structured ontology, which contains 5 modules and supports 5 languages. The 27 *entity-types* used as a seed set in our proposed ontology expansion approach are taken from the NyOn ontology itself.

The limitations of the above Indian legal IE systems and datasets are the small set of *entity-types* and relations, which together are referred to as the ontology. The number of *entity-types* is very less, which makes it harder to extract all the relevant legal named entities. Also, due to being fewer in number, they serve the purpose of a very broad domain. One example could be of type "Person", but in a legal context, "Person" could be anything among "Plaintiff", "Appellant", "Defendant", "Petitioner", "Respondent", etc. Labeling any of the instances with just "Person" is losing a lot of information. Hence, there is a critical need to expand the existing ontology, which can extract the maximum number of valid legal entities from the given legal document.

Chapter 3

Methodology

In this section we will discuss about the frameworks, FILOX (for ontology expansion) and FILKGC (for information extraction for KGC) in great detail, and in the Chapter 4 we will show their implementations and the results.

Pre-Processing

The pre-processing module is the common and crucial step for both frameworks.

In this module, first the judgments are scanned for any general or legal domain-specific contractions (Dr., Mr., etc.) and abbreviations (sec., IPC, etc.). The expansions are done based on the predefined abbreviations and contractions list which we have curated from legal books and the web. Secondly, we split these judgments by sentence level using Spacy's rule-based sentencizer - "en_core_web_sm". The pipeline is shown in Fig. 3.1.

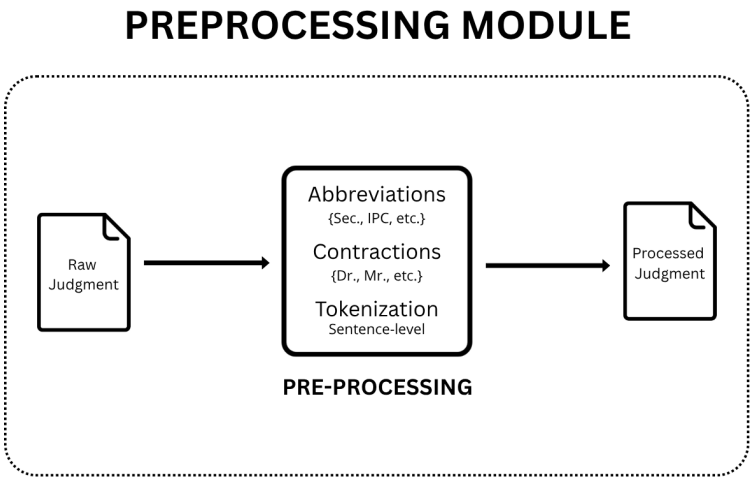

Figure 3.1: Judgment pre-processing pipeline.

3.1 FILOX: Framework for Indian Legal Ontology Expansion

3.1.1 Expanding *entity-types*

As shown in Fig. 3.2, the pipeline consists of the following 4 steps:

Step 1: We start with providing a seed set of 27 *entity-types* obtained from the NyOn [1] ontology in prompt 3.3. We pass the judgment sentence-wise in order to extract the legal entities and *entity-types* present in it. Paragraph-level context is also provided in the prompt to allow the LLM for better understanding and reasoning. We instruct the LLM to classify the entities present in the sentence with either the provided 27 *entity-types* or it can create new ones if none fit. This way, we keep the space for the LLM to expand the existing list.

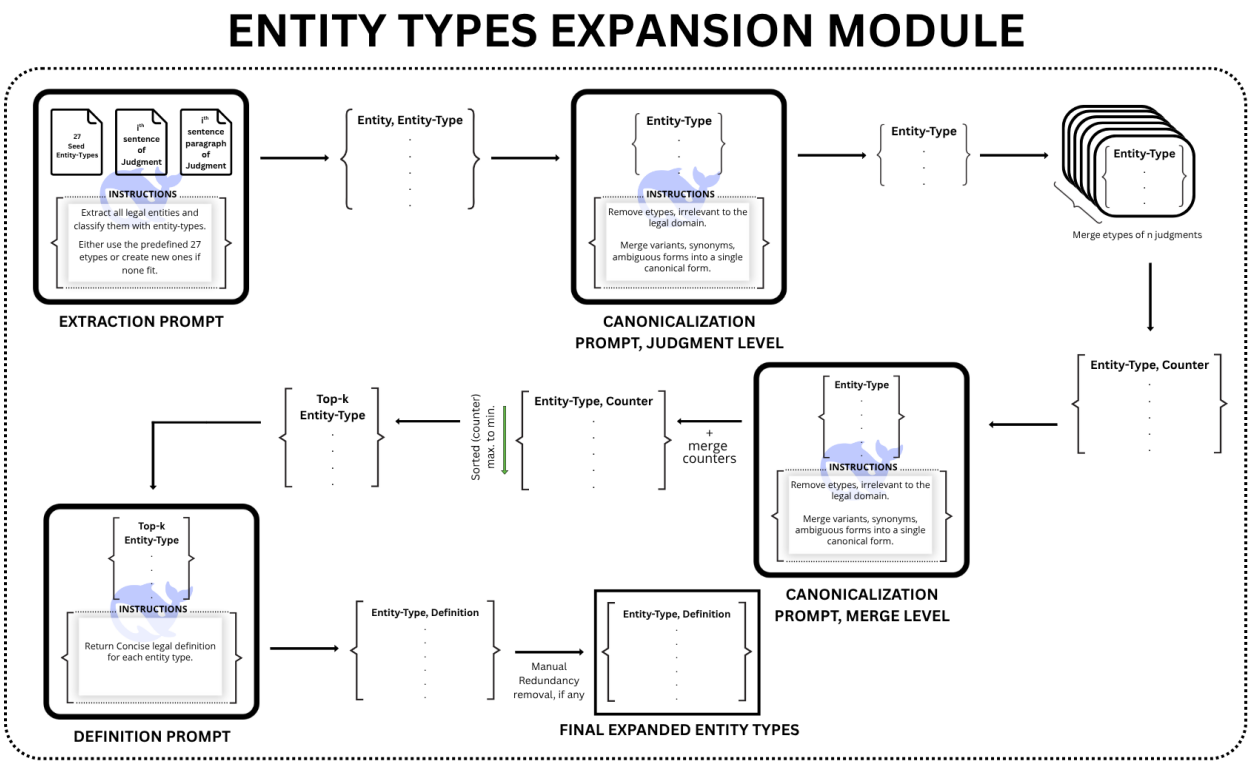

Figure 3.2: Enity-Types expansion pipeline.

For each *judgment* j_i, consisting of m sentences, we extract a set of entity and *entity type* pairs from each sentence. Let the set extracted from the s-th sentence of judgment j_i be denoted by:

$$E_{i,s} = \{(e^1_{i,s}, t^1_{i,s}), (e^2_{i,s}, t^2_{i,s}), \ldots, (e^{k_{i,s}}_{i,s}, t^{k_{i,s}}_{i,s})\}$$

where $k_{i,s} \geq 0$ indicates the number of extracted pairs from that sentence.

Legal Entity and Entity Type Extraction Prompt

You are a legal expert specializing in identifying legal entities in text.

Task:
Analyze the sentence below within its surrounding context. Identify and extract all entities from the sentence relevant to legal interpretation or analysis, assigning an appropriate entity type to each.

Output Format:
Only return a JSON array of objects, where each object includes exactly the following keys (all values as strings):
- entity — the entity instance
- entity_type — a label that best describes the type of entity

Use one of the predefined entity-types listed below. If none of them fits, you may define a new, clearly-named entity type.
Predefined entity-types:
{*entity_descriptions*}

Input:
Sentence: {*sentence*}
Context: {*context_text*}

Instructions:
- Output only a single JSON array.
- Do not include markdown, headings, comments, or explanatory text.
- Avoid any text outside of the JSON brackets.
- All JSON values must be enclosed in double quotes.
- No trailing commas or invalid formatting.
- Ensure entity-types are precise and consistent.
- Do not add notes in the output about newly generated entity-types.

Figure 3.3: Prompt for extracting entities and *entity-types*

The full set for the *entire judgment* j_i is obtained by merging the sentence-level sets:

$$E_i = \bigcup_{s=1}^{m} E_{i,s}$$

Step 2: We then perform the canonicalization of the *entity-types* on the judgment level using the prompt 3.4.

Canonicalization - Judgment level

Assume you are a legal expert who understands legal documents thoroughly.

Given the following list of entity-types, do the following:

1. Remove any entity-types not relevant to the legal domain.

2. Normalize/standardize the remaining types by merging naming variants into a canonical form.

3. Your output must be a single JSON object—starting with "{{" and ending with "}}", with no extra text or explanation outside the braces.

- *Each key in that object is a resolved (canonical) entity type (string).*
- *Each value is an array of the original variants (list of strings).*

Here is the list of entity-types: {entity_list}

IMPORTANT: *- Do not include any commentary, bullet points, or code fences. - Do not omit the outer curly braces.*

Figure 3.4: Prompt for canonicalizing *entity-types* at the judgment level

After the judgment-level canonicalization over n judgments, we get the corresponding *entity type* sets:

$$E'_n = \{E'_1, E'_2, \ldots, E'_n\}$$

where n judgments are defined as:

$$J_n = \{j_1, j_2, \ldots, j_n\}$$

Finally, the *master set* aggregating all canonicalized *entity type* pairs from all judgments is:

$$E = \bigcup_{i=1}^{n} E'_i$$

From this final master set E, we construct a new set T_c of *distinct entity-types* with their corresponding *counters*. Each element in this set is of the form:

$$T_c = \{(t_1, c_1), (t_2, c_2), \ldots, (t_p, c_p)\}$$

where t_j is a unique *entity type*, and $c_j \in [1, n]$ is the number of judgments in which type t_j was extracted.

Thus, the counter c_j reflects how widely an *entity type* t_j appears across the corpus of judgments. A value of $c_j = 1$ means that the type appeared in only one judgment, while $c_j = n$ indicates that it appeared in all the judgments.

Step 3: To remove the redundancies of the *entity-types* across different judgments, canonicalization on this new merged set T_c is needed one more time. So we do so by the prompt 3.5.

Postprocessing - Merge level

Assume you are a legal expert who understands legal documents thoroughly.

Given the following list of entity-types, do the following:

1. Remove any entity-types not relevant to the legal domain.

2. Normalize/standardize the remaining types by merging naming variants into a canonical form.

3. Your output must be a single JSON object—starting with "{{" and ending with "}}", with no extra text or explanation outside the braces.

- *Each key in that object is a resolved (canonical) entity type (string).*
- *Each value is an array of the original variants (list of strings).*

Here is the list of entity-types: {entity_list}

IMPORTANT: *- Do not include any commentary, bullet points, or code fences. - Do not omit the outer curly braces.*

Figure 3.5: Prompt for canonicalizing *entity-types* at merge level

During this canonicalization, we also club the counters of the merged *entity-types*, because it simply means the addition of counters of various versions of the same *entity-types* in different judgments. We then sort the list from highest to lowest counter value and select top-k *entity-types* from this set.

Step 4: This step aims at defining the selected top-k *entity-types*. For this we use the prompt 3.6.

Before freezing the final *entity-types* set, we manually delete redundancies, if any are left.

3.1.2 Expanding Relations

As shown in Fig. 3.7, the pipeline consists of the following 4 steps:

Entity-Type Definition Generation

You are a legal ontology specialist.

For each ENTITY_TYPE listed below, return exactly one JSON object of the form:
{"definition": "concise legal definition"}

Do NOT include any chain-of-thought, "Thinking...," commentary, bullet points, or extra text. Return only valid JSON and nothing else.

ENTITY_TYPE: {entity_type}

Figure 3.6: Prompt for defining *entity-types*

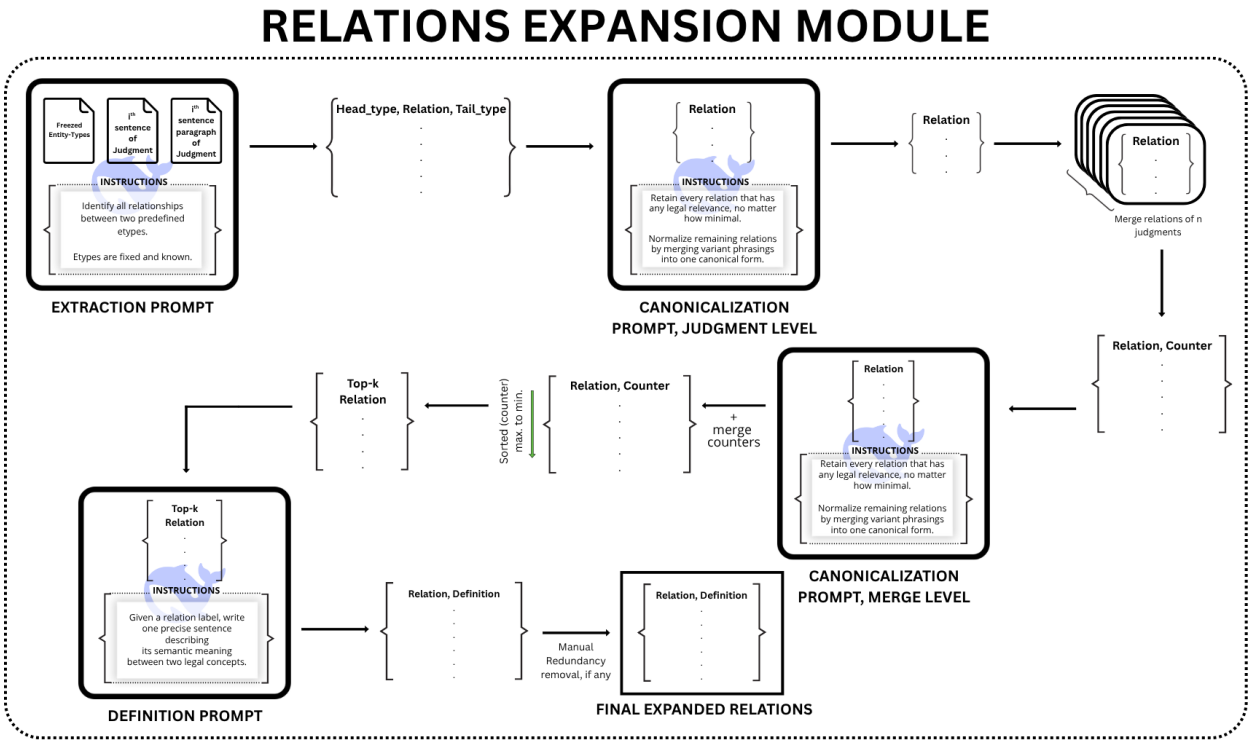

Figure 3.7: Relations expansion pipeline.

Step 1: To expand the relations list, we pass the list of predefined *entity-types* and their definitions, along with the judgment sentence-wise, the sentence's paragraph for the context, and instruct the LLM to identify the *entity-types* in that sentence from the predefined list and extract a valid legal relation between them.

The prompt for extracting *entity-types* and relationships between them is shown in Fig. 3.8.

Relation Extraction Prompt

You are a legal-domain relation-extraction expert.

Task:
Given a single sentence and its surrounding paragraph, identify all relationships between two predefined entity-types. The entity-types are fixed and known. Entities may overlap if they are involved in different relation pairs. Return all valid relations present in the sentence. If no valid relation exists, return an empty JSON array:

Output:
Return a JSON array, with each element being a JSON object containing exactly these string fields:

- *"head_type": string (the type of the first entity)*
- *"relation": string (the relation between them)*
- *"tail_type": string (the type of the second entity)*

Predefined entity-types and Definitions:
{*entity_descriptions*}

Input:
Sentence: "{*sentence*}*"*
Paragraph Context: "{*context_text*}*"*

Instructions:

- *Output only the JSON array, nothing else (no explanations, no markdown, no comments).*
- *Use double quotes for all JSON keys and string values.*
- *Ensure the JSON is valid (no trailing commas).*
- *If multiple relations exist, include each as a separate JSON object inside the array.*
- *If no relations are found, output an empty array:*

Figure 3.8: Prompt for extracting *entity-types* and relationship between them

For each *judgment* j_i, consisting of m sentences, we extract a set of relation triplets from each sentence. Let the set extracted from the s-th sentence of judgment j_i be denoted by:

$$R_{i,s} = \left\{ \left(e^1_{i,s},\ r^1_{i,s},\ e^{1'}_{i,s}\right),\ \left(e^2_{i,s},\ r^2_{i,s},\ e^{2'}_{i,s}\right),\ \ldots,\ \left(e^{\ell_{i,s}}_{i,s},\ r^{\ell_{i,s}}_{i,s},\ e^{\ell_{i,s}\prime}_{i,s}\right) \right\}$$

where $\ell_{i,s} \geq 0$ is the number of extracted triplets from sentence s.

The full set for the *entire judgment* j_i is obtained by merging the sentence-level triplet sets:

$$R_i = \bigcup_{s=1}^{m} R'_{i,s}$$

Step 2: Similar to *entity-types* judgment-level canonicalization, we perform for the relations, using the prompt 3.9.

Legal Relation Canonicalization Prompt - Judgment level

Assume you are a legal expert who understands legal documents thoroughly.

Task:

Given the following list of relations, do the following:

1. *Retain every relation that has any legal relevance, no matter how minimal; only exclude those that clearly fall outside the legal domain.*
2. *Normalize and standardize the remaining relation names by merging variant phrasings into a single canonical form.*
3. *Your output must be a single JSON object—starting with "{{" and ending with "}}", with no extra text or explanation outside the braces.*

Output Format:

- *Each key is a canonical relation name (string).*
- *Each value is a list of original relation strings (list of strings).*

Input:

{*relation_list*}

IMPORTANT:

- *Do not include any commentary, bullet points, or code fences.*
- *Do not omit the outer curly braces.*

Figure 3.9: Prompt for canonicalizing relations at judgment level

After the judgment-level canonicalization over n judgments, we get the corresponding *relations* sets:

$$R'_n = \{\, R'_1,\ R'_2,\ \ldots,\ R'_n \}$$

where n judgments are defined as:

$$J_n = \{j_1, j_2, \ldots, j_n\}$$

Finally, the *master relation-triplet set* aggregating all triplets from all judgments is:

$$R = \bigcup_{i=1}^{n} R_i$$

From this master set R, we construct a new set $\mathcal{C}_r$ of *distinct relations* with their corresponding *counts*. Each element of this set is of the form:

$$\mathcal{C}_r = \{ (r_1,\ c_1),\ (r_2,\ c_2),\ \ldots,\ (r_q,\ c_q) \}$$

where r_j is a unique relation label, and

$$c_j \in [1, n]$$

Legal Relation Canonicalization Prompt - Merge level

Assume you are a legal expert who understands legal documents thoroughly.

Task:

Given the following list of relations, do the following:

1. *Retain every relation that has any legal relevance, no matter how minimal; only exclude those that clearly fall outside the legal domain.*
2. *Normalize and standardize the remaining relation names by merging variant phrasings into a single canonical form.*
3. *Your output must be a single JSON object—starting with "{{" and ending with "}}", with no extra text or explanation outside the braces.*

Output Format:

- *Each key is a canonical relation name (string).*
- *Each value is a list of original relation strings (list of strings).*

Input:

{*relation_list*}

IMPORTANT:

- *Do not include any commentary, bullet points, or code fences.*
- *Do not omit the outer curly braces.*

Figure 3.10: Prompt for canonicalizing relations at merge level

is the number of judgments in which the relation r_j appears.

Thus, $c_j = 1$ indicates r_j occurred in exactly one judgment, while $c_j = n$ means it was present in all n judgments.

Step 3: To remove the redundancies of the relations across different judgments, canonicalization on this new merged set $\mathcal{C}_r$ is needed one more time. So we do so by the prompt 3.10.

During this canonicalization, we also club the counters of the merged relations, because it simply means the addition of counters of various versions of the same relation in different judgments. We then sort the list from highest to lowest counter value and select top-k relations from this set.

Step 4: This step aims at defining the selected top-k relations. For this we use the prompt 3.11.

Legal Relation Definition Prompt

You are a legal expert. Given a relation label, write one precise sentence describing its semantic meaning between two legal concepts.

Output Format:
A single JSON object with exactly two keys:
- *"relation": the input label (string)*
- *"definition": a single concise sentence (string)*

Input:
Relation: {relation}

IMPORTANT:
- *Do not include any labels, bullet points, or extra commentary—only the sentence itself.*
- *Produce exactly one JSON object per invocation, with no surrounding text or fences.*

Figure 3.11: Prompt for defining relations

Before freezing the final relation set, we manually delete redundancies, if any are left.

3.2 FILKGC: Framework for Indian Legal Knowledge Graph Construction

This framework is aimed at extracting information in the form of triplets for constructing KGs by utilizing LLMs. The framework uses ontology created from the FILOX framework.

As shown in Fig. 3.13, the pipeline consists of the following 3 steps:

Step 1: The prompt 3.12—constructed using the pre-processed sentence, paragraph context, few-shot examples, predefined legal *entity-types*, and instructions via In-Context Learning (ICL) and persona-based (legal expert) prompting—is sent to the locally hosted DeepSeek-R1 server. It returns a .json file containing 'head, head type, relation, tail, tail type, definition' *(h, ht, r, t, tt, d)*. Here, 'head' and 'tail' are the entities extracted by the LLM, 'head type' and 'tail type' are their corresponding predefined *entity-types*, 'relation' connects them, and 'definition' provides a one-line description of the triplet *(h, r, t)*.

Legal Entity & Relation Extraction Prompt

Instructions:
You are a legal expert. Extract all entities and legal relations from the sentence below, given the paragraph context, using only the following entity-types exactly as defined.

entity-types and Definitions:
{*entity_descriptions*}

Few-Shot Examples (for guidance):
{*few_shot_examples*}

For each relation, return a JSON object with exactly these keys (all values must be strings):

- `head`*: the first entity text*
- `head_type`*: one of the provided entity-types*
- `relation`*: the phrase linking head and tail*
- `tail`*: the second entity text*
- `tail_type`*: one of the provided entity-types*
- `relation_definition`*: a brief explanation of the legal connection*
- `sentence`*: the full sentence text*
- `sentence_number`*: the sentence's line number*

Output Format:
Output only a single valid JSON array. Do not include any extra text, markdown formatting, or explanations.

Input:
Paragraph_context: {*context_text*}
Sentence: {*sentence*}
Sentence_number: {*sentence_number*}

Figure 3.12: Prompt for extracting triplets

Step 2:: However, the extracted entities are ambiguous and redundant due to the inherent generative nature of how these LLMs work. Due to this, 'Tata' and 'Ratan Tata' are considered as two distinct entities. To resolve this, we create vector embeddings of all the extracted entities and then perform agglomerative clustering on the basis of their cosine similarity. This way, we create 'n' number of clusters, each having semantically similar entities as their members. Each member is then assigned a canonical name, which is the largest string of the cluster. This way, we add 2 new attributes *(ch, ct)* to the list *(h, ht, r, t, tt, d)*, making it *(h, ch, ht, r, t, ct, tt, d)*. 'ch' and 'ct' represent canonical head and canonical tail, respectively.

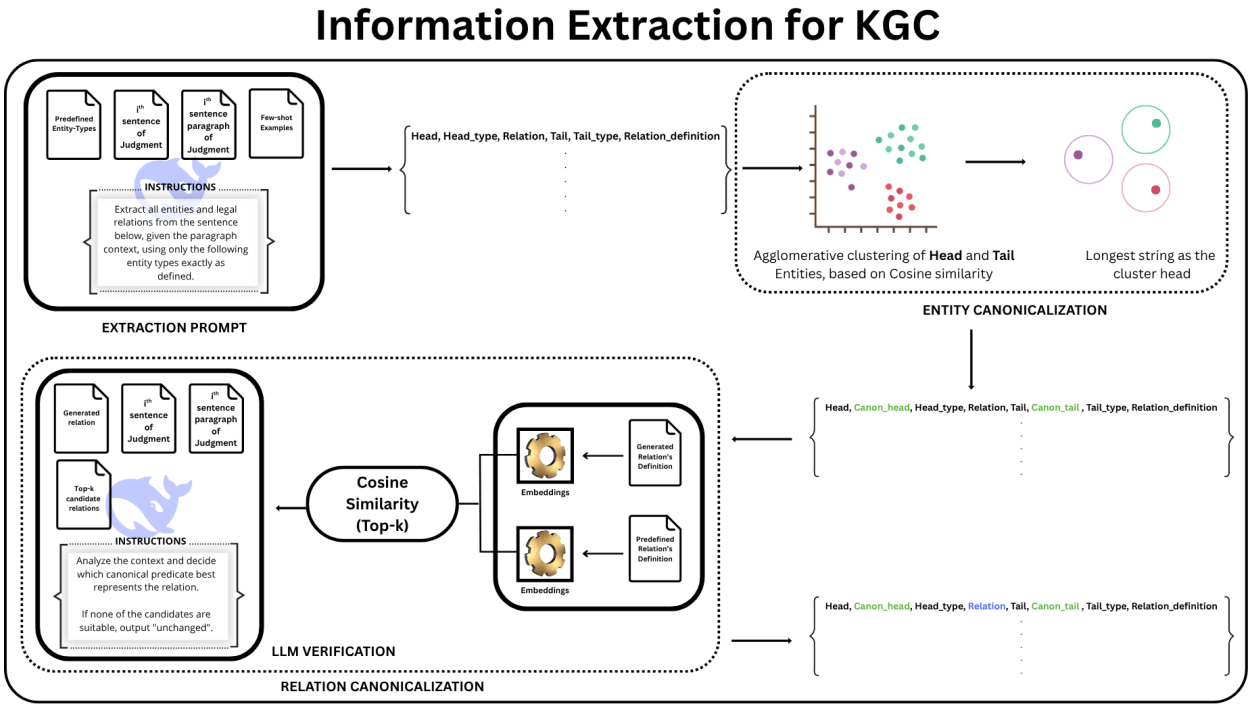

Figure 3.13: Information Extraction for KGC.

Step 3: The extracted relations are also ambiguous and redundant because of producing synonyms at various places for representing a single relation. For example, for the sentence:"The court ruled that ABC Corp breached the contract with John Doe", LLM may produce three triplets: (ABC Corp, VIOLATED, contract), (ABC Corp, FAILED_TO_HONOR, contract), and (ABC Corp, DID_NOT_UPHOLD, contract). They all are referring to the same relation, but due to different word selection, they are treated as distinct. We resolve this in two stages: Stage 1: We create vector embeddings of the definitions 'd' from the set *(h, ch, ht, r, t, ct, tt, d)*, defining the entities and relations. Then we create vector embeddings of the definitions 'd′' of the pre-defined set of legal relations. We compute cosine similarity and select the top k (5) relations from the pre-defined set that are similar to the extracted relation 'r'. In stage 2, we send these top-k relations, the original relation 'r', the original sentence, the sentence paragraph , and the head and tail entities for LLM

verification. The intuition behind this is we do not forcefully, unknowingly, or rationally want to replace the current extracted relation 'r' with the most similar relation from the pre-defined set, even if it does not make any sense after the replacement in the triplet. So, the idea is to select the top-k similar options and then ask the LLM for its validity in the triplet after replacing it with each of the k choices, and if the LLM does not think that any of it can do justice to the semantic meaning of the triplet, it keeps it unchanged. This way we have 'r′', the final relation in our set, *(h, ch, ht, r', t, ct, tt, d)*.

The prompt used for LLM verification is given in 3.14:

LLM Verification Prompt

You are a legal expert. You are provided with:
1. An original sentence from a legal document.
2. The current relation (predicate) extracted from that sentence.
3. The paragraph of the sentence, for better context.
4. A candidate list of canonical predicates along with their definitions.
Your task is to analyze the context and decide which canonical predicate best represents the relation. If none of the candidates are suitable, output `"unchanged"` *to indicate that the predicate should remain as-is.*

IMPORTANT: *Output strictly valid JSON. Do not include any chain-of-thought or commentary. The output must be a JSON object mapping the current relation (as provided below) to its chosen canonical predicate or the string* `"unchanged"`*.*

Input:
Original Sentence: {*original_sentence*}
Current Relation: {*current_relation*}
Relation Definition: {*relation_definition*}
Original Sentence Paragraph: {*sentence_paragraph*}

Output:
Revised Result:

Figure 3.14: Prompt for performing LLM verification for relation canonicalization

These high-quality extracted attributes can be used to create and query the knowledge graph built specifically for the legal domain.

Chapter 4

Implementation and Experimental Results

4.1 Implementation Details

We implement FILOX and FILKGC using the DeepSeek-R1 14B parameter model by downloading it and running it on the Omni-Layer Learning Language Acquisition Model, 'Ollama', which is an open-source tool designed for running and managing large language models (LLMs) locally on the computer/server. Details about the hardware configuration is mentioned in the Table 4.1.

Component	**Details**
RAM	512 GB
Processor	Intel® Xeon® Gold 6326 CPU @ 2.90 GHz (2 processors)
OS Type	64-bit operating system, x64-based processor
OS Name	Windows Server 2022 Standard
GPU	NVIDIA A100
VRAM	80 GB
Python Version	3.12.3

Table 4.1: Hardware Configuration of the server

4.1.1 Ollama Context Window Lengths: 8k vs 32k

The intuition behind this experiment was that although the context length of all the DeepSeek models is 128k tokens, to make it more efficient for running on smaller devices, Ollama further caps this limit to 8k tokens by default. Increasing the context window allows the model to consider more context, improving the coherence of longer texts. However, it also requires more computational resources. If we reduce this value, the generation process

can be sped up but might result in less coherent or contextually aware outputs for longer texts. As we were using the In-Context Learning (ICL) technique, we were having longer prompts, which included many *entity-types* and their definitions in the prompt along with the rest of the components, and hence a longer context for the LLM was needed. So we increased it to 32k tokens to avoid such scenarios where the prompt might get trimmed by the Ollama just because it was too large to fit in the allowed context window limit.

We have seen major improvement after the change. However, the utilization of VRAM increased drastically from $\tilde{1}$1GB in the 8k token limit to $\tilde{5}$8GB in the 32k token limit.

4.1.2 Dataset(s)

We collected a set of 114 publicly available Supreme Court judgments, obtained from the offficial Supreme Court of India [1] website. These belong to the spectrum of legal subdomains (e.g., civil, criminal, etc.). More details about the dataset are provided in Fig. 4.2.

Metric	**Value**
Total # judgments	114
Total # words	5,59,034
Total # sentences	18,071
Average # words per judgment	4,903.8
Average # sentences per judgment	158.51

Table 4.2: Statistics of the judgment dataset

4.2 Implementation

4.2.1 Expanding Indian Legal Ontology Using FILOX

Expanding *entity-types*

We first preprocessed all the judgments using the preprocessing pipeline as explained in 3.1.1. Then we passed them into the first phase of the framework. We extracted a total of 1,427 *entity-types*. The histogram in Fig. 4.1 shows the count of 1,274 *entity-types* on the x-axis and the number of judgments in which they appeared (out of 114) on the y-axis. The right-skewed distribution of the histogram demonstrates a large number of *entity-types* that appeared in only 1 judgment, and very few appeared more frequently across the judgments. After performing judgment-level canonicalization, we were left with 674 *entity-types*. During this step, redundant *entity-types* were removed such as, (Date and Time, date_amount, Date_and_Time, date_time, DateAndTime, Dates, dates_and_times, DateTime, Date-Time,

[1] https://www.sci.gov.in/

etc.) were replaced with 'Date and Time'. There were also some *entity-types* which were irrelevant to the legal domain. For example (Mental Capacity, train, etc.). We simply removed them from the list. Then we merged the canonicalized results of all the judgments and performed merge-level canonicalization, after which we were left with 587 *entity-types*. From this list, we selected the top 100 *entity-types*. Then we generated the definitions of these 100 *entity-types*. Before freezing the final set, we found 13 redundant types, such as (date_time, date_and_time), (legislature_body, legislature), (judge, court_official_judge), etc.. After removing them, we were left with the final set of **87 unique *entity-types***. The final set of *entity-types* is listed in table - 4.3.

The framework is scalable, and one can easily scale the ontology using it to the broader level, depending on the resources they may have. One can also use the created set of 87 *entity-types* directly in their pipeline.

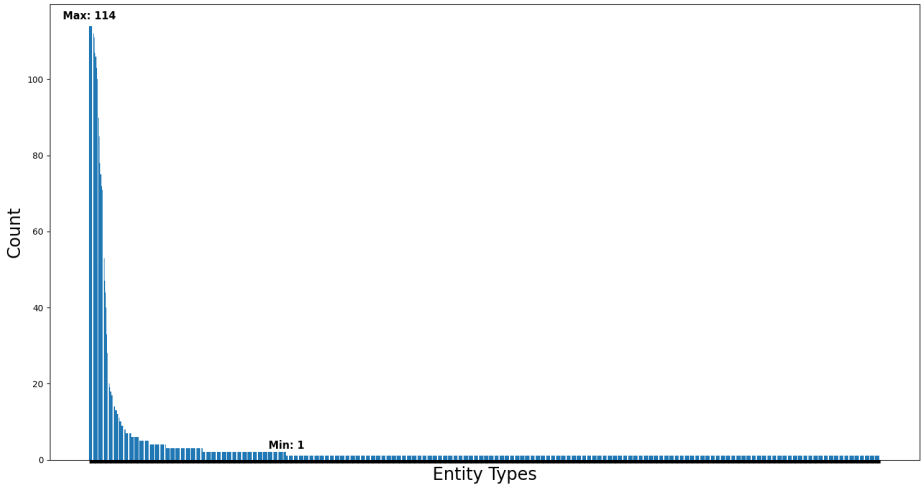

Figure 4.1: Distribution of the count of judgments (y-axis) in which the *entity-types* (x-axis) occur at least once

Table 4.3: Expanded *entity-types*, arranged in alphabetical order (row-wise)

accused	act	action	agent
appeal	appellant	application	argument
asset	authority	beneficiary	business
case	charge	citation	claim
company	complainant	constitution	contract
conviction	court	courtofficial	bench
crime	date	dateandtime	deceased
decision	definition	doctrine	document
estate	event	evidence	expertwitness
fact	financialterm	goods	government
governmentagency	governmentofficial	judge	jurisdiction
learnedcounsel	lawyer	legalauthority	legalconcept
legaldoctrine	legaldocument	legalprinciple	legalproceeding
legislature	location	monetaryvalue	obiterdictum
offense	official	order	organization
party	person	petition	petitioner
plaintiff	police	policy	position
procedure	process	property	prosecution
provision	publication	ratiodecidendi	regulation
respondent	role	rule	section
sentence	statute	suspect	tax
victim	weapon	witness	

Expanding Relations

The whole process of expanding relations is similar to the process of *entity-type* expansion, just the stats are different. So we started with the first phase, i.e extraction, in which we extracted 3,669 relations. The histogram in Fig. 4.2 shows the count of 3,669 relations on the x-axis and the number of judgments in which they appeared (out of 114) on the y-axis. The right-skewed distribution of the histogram demonstrates a large number of *entity-types* that appeared in only 1 judgment, and very few appeared more frequently across the judgments. After performing judgment-level canonicalization, we were left with 1,118 relations. The redundant examples included (accused, accused_in_case, accused_of),(testify, testified, testifies, testifies_against, testifies_in, testifying_in), etc.. After resolving these, we merged the relations of all judgments and performed merge-level canonicalization, but passing 1,118 relations at once in a prompt was confusing the LLM; hence, we passed the chunk of 100 relations in each iteration and combined the results of all the iterations, after

which we were left with 626 relations.

We further canonicalized these 626 relations, but all at once this time, to resolve the redundancy across the prior chunks of 100 relations each. After this, we were left with 602 relations. We then sorted the list based on the highest to lowest counter value and selected the top **320 relations** for the final set. The final relations are listed in Tables - 4.4, 4.5, 4.6.

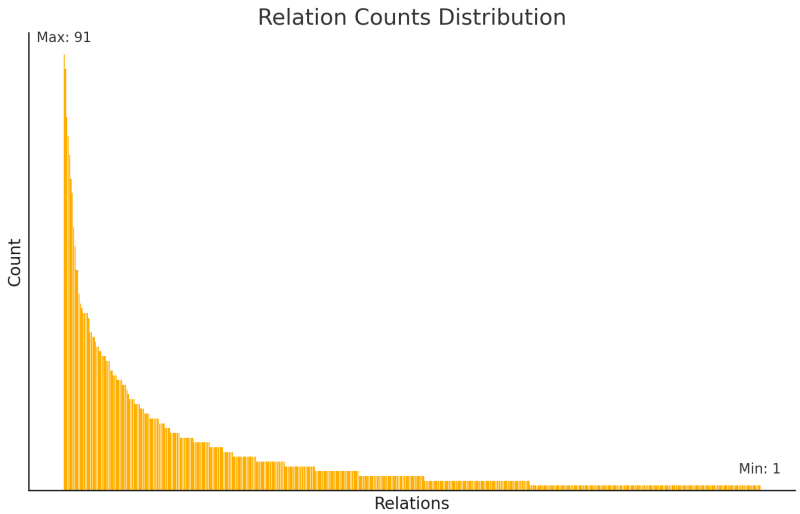

Figure 4.2: Distribution of the count of judgments (y-axis) in which the relations (x-axis) occur at least once

Table 4.4: Final Legal Relations (Part 1)

acceptance	accompany	accuse	accused
accusing	acquire	acquitted	act
act_under	action	acts_on_behalf_of	addressed
adjust	admit	advanced	affect
affiliate	affirmation	afford	against
agency	agent	agreement	allegation
allowance	analysis	answer	appeal
appeal	arrest	assault	assess
assessment	asset	assign	assist
associate	authority	authorityover	authorize
authorize	award	based_on	binding
breach	bring	calculate	call
cancel	cannot_claim	carried	carries
case	case_related	caught	cause
certainty_of_fact	challenge	charge	citation
claim	classification	close	commission
commit	communication	complaint	compliance
conclusion	conduct	confer	confine
confirm	conflict	conveyance	convicted
conviction	corroboration	court_proceedings	creation
crime	custody	date	dealing_with
dealt_with	death	decide	decision
declare	defence	defendant	define
delegate	deliver	demand	deny
depose	deprive	designation	determination
dictate	did_not_commit	direct	disallow
discharge	disclose	discriminate	dismiss
dispose_of_case	disregard	distribution	does_not_alter
does_not_apply_to	does_not_have	does_not_impose_obligation	does_not_prohibit
does_not_require	domicile	duty	duty_imposed_by
effect	effective_date	eject	eligibility
enactment	encroachment	enforcement	engagement
entitled_to	evidence	evidence_for	examination
examination	exclude	exclusion	executed
exercise	express_opinion	extend	facilitate

Table 4.5: Final Legal Relations (Part 2)

failure	falls_under	false	fault
file	find	form	frame
hears	heir	held_that	impose_penalty
imposes_restrictions_on	imposes_tax	incorporate_law	influence_legal
information_disclosure	initiate_action	injury_claim	institution_legal
intent	intent_knowledge	interaction	interest
interest_in	interpretation	intervention	introduction
invalidation	investigation	investment	invitation
involved_in	involvement	is	issued
issuing	judge	judgment	judicial_authority
jurisdiction	kill	knowledge	liability
maintain	member_of	mention_in_document	modify
monetary_value	non_compliance	notice	obligate
obligation	observe	offer	omission
on	operate	opinion	oppose
order	originate_from	outside	overrule
oversaw	oversees	owe	own
paid	part	participant	parties
party	payment	penalty	perform
permit	petition	place	position
possess	power	prescribed	present
procedure	produce	prohibit	property
prosecution	protect	provide	provision
publish	purchase	pursuance	quash
quote	raise_before_court	receive	record
recover	reduce	refer	refuse
regard	registered	regulated	reject
relate	release_on_bail	reliance	remove
rendered_judgment	repeal	replace	report
represent	representation	reprimand	request
require	response_obligation	restriction	result
review	right	role	rule
sanction	save	scrutiny	search

Table 4.6: Final Legal Relations (Part 3)

section	secure	seek	sentence
serve	shall	show	sign
similar	slander	state	status
strike_action	subject	submit	succession_rights
sue	supersede	supervise	supplementary
support	suspect	suspend	tax
terminate	testify	testimony	transfer
transfer	treat	trial	urge
use	validate	validation	value
versus	vested	victim	violate
visit	was_allowed	was_charged	was_conducted_by
was_in	was_involved_in	was_made_by	witness_relation

4.2.2 Extracting Information for KGC using FILKGC

There are a total of **36,070** triplets extracted from 114 judgments using FILKGC's extraction phase, which makes an average of 316 triplets/judgment. These consist of 33,790 distinct entities (including heads and tails), and 21,629 relations. Unfortunately, we could not proceed with entity canonicalization and relation canonicalization for all the 114 judgments, due to time and compute constraints, but the pipeline has already been built and tested with a single judgment. We expect the reduction in the number of both entities and relations after the redundancy removal and merging of variants. After performing the said two steps, a rich KG can be built around the dataset of 114 judgments.

Comparison Between 5 Methods of Judgment Context Windows

We experimented with the 5 different context windows passed in the prompt. Our motive behind this was to know which context window extracts the maximum number of distinct entities. There are many complex inter-sentence references found in judgments, which are hard to interpret on just sentence-level context; it requires additional context to understand the exact meaning of the scenario. Also, to know the impact on 1:1 and 1:Many entities. 1:1 here refers to 1 entity classified as only 1 *entity-type* throughout the whole judgment, whereas 1:Many refers to 1 entity classified as more than 1 *entity type* throughout the whole judgment, depending on the context given in the prompt. Examples of 1:1 entities include Arun Kumar: Petitioner, and for 1:Many include Arun Kumar: Petitioner, Appellant, Individual, Person. We expected the decrease in count of 1:Many entities as the context window gets bigger and bigger, reason being that LLM have more context to decide and assign the *entity-type* to the extracted entity, and hence is less delusional about its choices.

The different context windows are:

1. sentence only - no context provided, other than just passing the sentence
2. paragraph ±0 - paragraph containing that sentence is provided as context
3. chunk ±1 - a total of three paragraphs are provided as context: one paragraph containing that sentence, one paragraph above it, and one paragraph below it.
4. chunk ±2 - a total of five paragraphs are provided as context: one paragraph containing that sentence, two paragraphs above it, and two paragraphs below it.
5. chunk ±3 - a total of seven paragraphs are provided as context: one paragraph containing that sentence, three paragraphs above it, and three paragraphs below it.

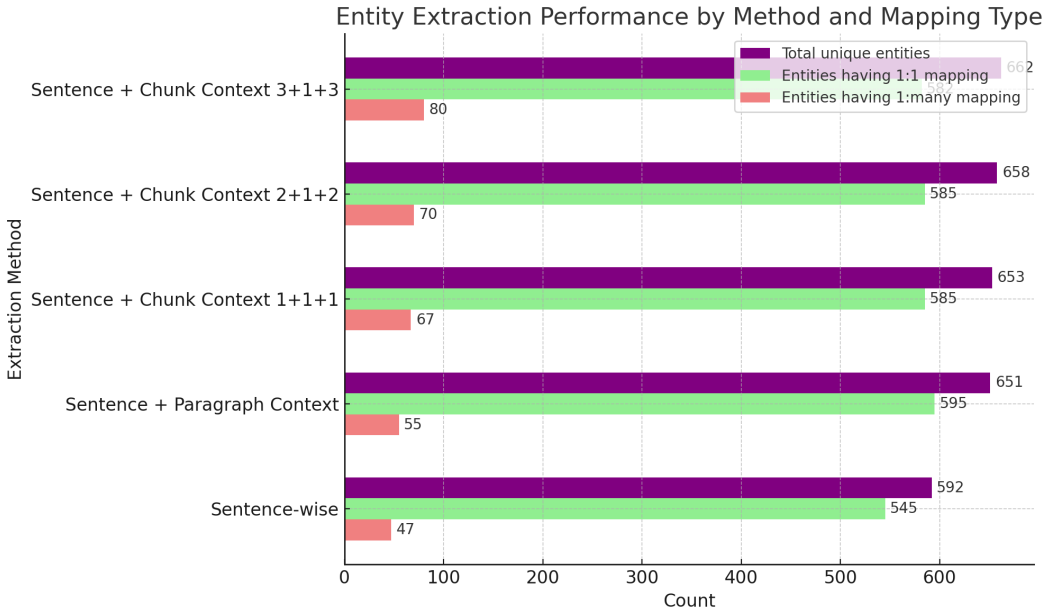

Figure 4.3: Distribution of the five context window lengths (y-axis) against the counts of unique entities, entities with 1:1 mappings, and entities with 1:many mappings (x-axis)

In all the above methods, as shown in Fig. 4.3, the entities are extracted from the sentences only, just for the additional context for the LLM; more information in the form of paragraphs is added in the prompt.

But to judge the best method among 5 on the basis of maximum number of unique entities, there is the need for entity resolution/linking. To understand it better, let's take an example of the sentence - "The said two petitions were filed by the respondents herein under section 482 of Criminal Procedure Code ('Code' for short)." From this sentence, the LLM extracts the following entities: 'section 482', 'section 482 of Criminal Procedure Code', and 'section 482 of Criminal Procedure Code ('Code' for short)'. Now, if we count these as total unique entities, these three would be considered as three distinct entities; however, they

are not. Hence, to judge the methods fairly and to remove redundancy from the extracted entities, we perform the entity resolution as mentioned in the methodology section 3.2, step 2.

Embedding Models For Entity Canonicalization

To perform the above discussed entity canonicalization, we chose to use the embedding method for it. However, to create vector embeddings of the entities, we experimented with 5 different embedding models. More information about the models can be seen in table 4.7. We selected the best method on the basis of human evaluation after keeping cosine similarity for all at 0.9. From this experiment, Sentence-transformers/all-MiniLM-L6-v2, formed the best clusters among all, after which we tweaked the values a bit more and found 0.85 most suitable. The comparison file link is attached in the footnote [2] .

S.No.	Model Name	# Parameters	Details
1	nlpaueb/legal-bert-base-uncased	∼110M	Finetuned on European court data
2	nlpaueb/legal-bert-small-uncased	∼42M	Finetuned on European court data
3	bhavyagiri/InLegal-Sbert	∼110M	Finetuned for Indian legal domain
4	Stern5497/Sbert-legal-xlm-roberta-base	∼125M	Finetuned on Legal data
5	Sentence-transformers/all-MiniLM-L6-v2	∼22M	Fast, Good, for general semantics

Table 4.7: Choosing embedding models

Comparison Between 5 Methods of Judgment Context Windows, after entity canonicalization

After allotting each entity its canonical name, we again compared the 5 methods (sentence, paragraph ±0, chunk ±1, chunk ±2, chunk ±3) on the basis of total unique canonical names, instead of total unique entities (which were redundant). As shown in Fig. 4.4 and Fig. 4.5, the purple bar shows the total unique entities in both the figures, and paragraph ±0 method is consistent in both the scenarios before and after the canonicalization, it extracts the maximum number of entities after chunk ±3 method. Also, the red bar, which represents the count of 1:Many entities, also decreased after entity resolution. From this ablation study, we found out that including paragraph as the context along with the sentence in the prompt extracts the maximum number of unique entities.

[2] https://docs.google.com/spreadsheets/d/1VARJDnrdeiXOmUfmRX9uSgAeCxA35jyJ_pMHNlGctpc/

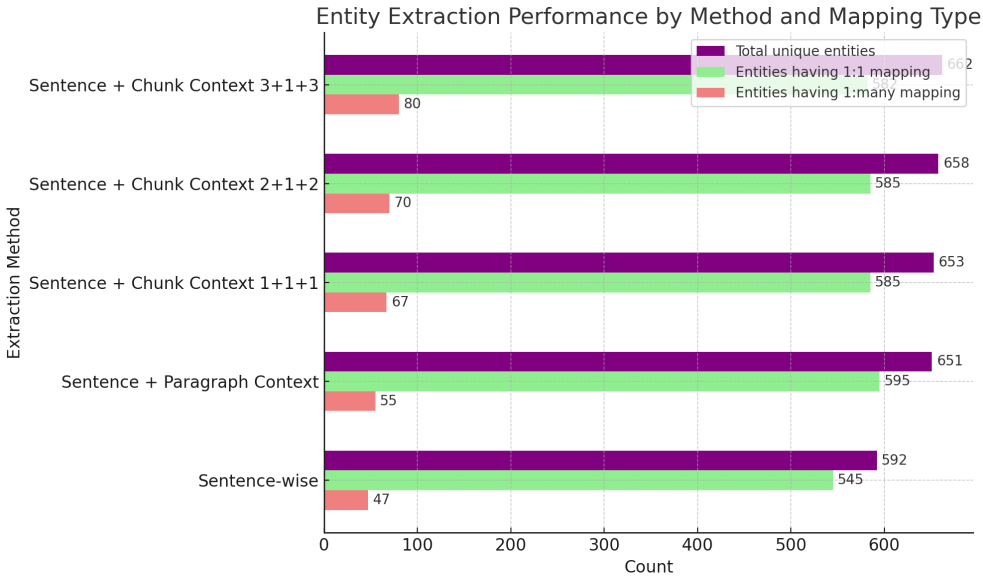

Figure 4.4: Distribution of the five context window lengths (y-axis) against the counts of unique entities, entities with 1:1 mappings, and entities with 1:many mappings (x-axis) **(Before Canonicalization)**

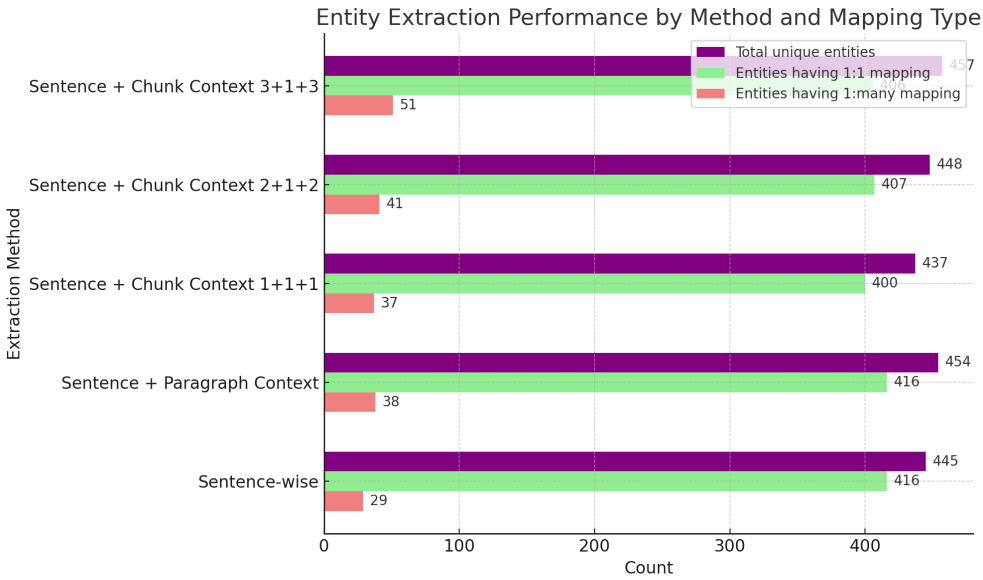

Figure 4.5: Distribution of the five context window lengths (y-axis) against the counts of unique entities, entities with 1:1 mappings, and entities with 1:many mappings (x-axis) **(After Canonicalization)**

Chapter 5

Conclusion and Future Work

In this project, we have developed the LLM-based frameworks FILOX and FILKGC for Indian legal ontology expansion and knowledge graph construction, respectively. We have used the DeepSeek-R1 14b model as the Large Language Model (LLM) for this project. Using FILOX, we have expanded the ontology to **87 legal *entity-types*** and **320 legal relations**. Similarly, using FILKGC, we have extracted the information in the form of triplets from 114 Indian legal judgments using the expanded ontology. The count of **unique triplets** from all these judgments is **36,070**.

As these frameworks try to fill the gap in present IE research, there are a few weak areas that need to be worked upon. To add few, firstly, there are the high computational requirements of LLMs to run locally or high server costs, in-case using API's. Secondly, the addition of noise in the output, which makes it hard to extract information from the particular format. This was very poor in previous generation models like LLama's 1 and 2 series, and Mistral's 7b series and is getting better nowadays, but still it makes the job hard to instruct the model to follow a specific output structure without making silly mistakes. Also, the models fine-tuned on specific tasks using the data annotated by experts still perform best in the majority of the cases compared to the LLM prompting techniques. But, in cases where there is a lack of expert-annotated datasets, LLM-based prompting techniques work as a viable alternative, offering competitive performance without the need for extensive manual annotation, especially when combined with few-shot examples, persona-based prompting, and in-context learning (ICL).

This pipeline can be scaled to more numbers of judgments as future work so as to create a big database of triplets of Indian legal judgments. Also, the entity and relation canonicalization could be performed as future work, which was left incomplete due to time and compute constraints in this project, as part of the IE for KGC.

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Chapter 6

Appendix-1

Directory Structure of `legalkg_llm` [1] Repository:

```
.vscode/
    |- settings.json
datasets/
    |- judgements/
    |   |- 1953_24.txt
    |   |- ...
    |   |- 2018_C_7.txt
    |- abbreviations.json
    |- contraction_map.txt
    |- ...
    |- relations.txt
output/
    |- drive_link.md
prompts/
    |- etype_postprocessing.txt
    |- ...
    |- relations_definition.txt
scripts/
    |- filkgc/
    |   |- entities_and_relations.py
    |- filox/
        |- entity_types/
        |   |- entity_relation_extraction.py
        |   |- ...
        |   |- preprocessing.py
        |- relations/
```

[1] https://github.com/Harsh-Yadav-02/legalkg_llm

```
        |   |- expansion_relations.py
        |   |- ...
        |   |- relations_definition.py
        |- dataset_analysis.py
.gitignore
output.log
requirements.txt
```

The expanded *entity-types* with their definitions are provided in tables from the next page.

S.No	Entity Type	Definition
1	accused	The individual legally charged with committing a crime, against whom prosecution is being pursued.
2	act	A formal written law passed by a legislature.
3	action	A legal claim or cause of action recognized by law that enables an individual or entity to seek redress or remedy through judicial proceedings for an injury or wrong committed.
4	agent	A legal entity designated to act on behalf of another person or organization, having the authority to make decisions or perform actions that legally bind the principal.
5	appeal	A formal request made to a higher court or tribunal to review and overturn a lower court's decision.
6	appellant	A party who brings an appeal, challenging a lower court's decision as incorrect or unlawful.
7	application	A formal request made to a court, tribunal, or other legal authority for a specific action or benefit.
8	argument	A statement or series of statements made to establish or prove a point of law, typically presented during legal proceedings such as trials or hearings.
9	asset	A legally recognized item, property, or right that is owned by an individual, business, or entity and has value in terms of money, benefits, or other resources. It includes tangible items (e.g., real estate, vehicles) and intangible items (e.g., intellectual property, shares).
10	authority	The right or power to make decisions, enforce rules, or take action within a legal framework.
11	beneficiary	A person legally entitled to receive benefits or advantages from another, typically under a trust, will, insurance policy, or other legal arrangement.
12	business	An organization engaged in commercial, industrial, or professional activities, typically aiming to make a profit.
13	case	A legal proceeding in which one party seeks to enforce rights against another party through a court or tribunal.
14	charge	A formal accusation in criminal law alleging that an individual has committed a specific offense.
15	citation	A reference to an authority or source cited as evidence or justification for a statement or proposition.

S.No	Entity Type	Definition
16	claim	A statement or assertion that affirms a party's legal right to something.
17	company	An entity formed under legal jurisdiction to engage in business activities, consisting of individuals organized for collective action.
18	complainant	A person who files a formal complaint in a legal proceeding.
19	constitution	A fundamental law establishing the principles, rules, and structure of a government or organization, outlining the rights, duties, and limitations of its entities and members.
20	contract	A legally binding agreement between two or more parties that outlines mutual obligations and rights, requiring consideration, mutual assent, capacity, and a lawful purpose.
21	conviction	A formal determination of guilt in a criminal proceeding.
22	court	A judicial body established by law or tradition to adjudicate disputes, resolve legal issues according to applicable laws, and enforce justice.
23	courtofficial	A person appointed to manage and oversee the administrative and operational functions of a court.
24	bench	A panel of judges who has the authority to preside over legal proceedings and render judgments.
25	crime	An act or omission that violates a statutory law and is subject to criminal penalties.
26	date	A specific point in time, usually represented as a day, month, and year, used to record events or establish deadlines, durations, or obligations.
27	dateandtime	A specific point in chronological order, consisting of both date (day, month, year) and time (hours, minutes), used to record when events occur or obligations are due.
28	deceased	A person who has died.
29	decision	A formal and official resolution or determination reached by a court or tribunal on a legal matter.
30	definition	A concise explanation of the meaning or significance of a legal term or concept.

S.No	Entity Type	Definition
31	doctrine	A system of established legal principles, rules, or theories that provide guidance on interpreting and applying laws.
32	document	A written or printed instrument containing information relevant to legal matters.
33	estate	All legal and equitable interests in real and personal property, including future interests, that an individual owns at the time of their death.
34	event	An occurrence, happening, or incident with legal relevance or significance.
35	evidence	Proof presented in a legal proceeding to establish a fact or support an assertion.
36	expertwitness	A person with specialized knowledge or expertise in a particular field qualified by the court to provide testimony, analysis, or opinion that assists the judge or jury in understanding complex issues.
37	fact	A specific event or circumstance that is verifiably true and serves as the foundation for legal reasoning or decisions.
38	financialterm	A word, phrase, or abbreviation used within finance and law that has an established, recognized meaning or definition.
39	goods	Tangible movable personal property.
40	government	The system by which a state or country is governed, including its institutions and branches that create and enforce laws.
41	governmentagency	An organization established by law or governmental authority to perform specific functions such as administering programs, enforcing laws, regulations, policies, and standards.
42	governmentofficial	An individual appointed or elected to serve in a public office, exercising authority and performing specific governmental functions.
43	judge	A judicial officer appointed to hear and decide legal cases in a court of law.
44	jurisdiction	The authority of a court to hear and decide cases involving a particular subject matter or geographic area.
45	lawyer	An individual licensed to practice law who provides legal advice, representation, and advocacy for clients in court or other legal proceedings.

S.No	Entity Type	Definition
46	learnedcounsel	A senior or distinguished lawyer who has demonstrated significant expertise and experience in legal practice.
47	legalauthority	The power or right granted by law to an individual, organization, or entity to act in a specific manner within the legal framework.
48	legalconcept	A fundamental principle, rule, or idea that forms the basis of legal systems and frameworks.
49	legaldoctrine	A body of law principles and rules established through judicial decisions and precedents guiding the application and interpretation of legal systems.
50	legaldocument	A written or electronic record containing legal information, instructions, or agreements.
51	legalprinciple	A fundamental rule or guideline that underpins and guides the interpretation and application of laws across jurisdictions.
52	legalproceeding	A formal process conducted under judicial authority to enforce a law, regulation, or principle.
53	legislature	A body of representatives elected to create, pass, and implement laws.
54	location	A specific place or area recognized by law and having legal significance and functions.
55	monetaryvalue	The financial worth or value of an entity, asset, or right, expressed in monetary terms and determined according to applicable laws, regulations, or agreements.
56	obiterdictum	A remark or observation made by a judge during a court proceeding that is not necessary to the resolution of the issues before the court and therefore not considered binding precedent.
57	offense	A violation of the law that results in a penalty or sanction.
58	official	An entity officially recognized or established under law, such as a government agency, public institution, or authority.
59	order	A directive or command, often issued by a court or other authority, requiring specific actions to be performed or behaviors to be adopted.
60	organization	An organized body or group established to achieve specific objectives, characterized by a structured system of roles and processes.

S.No	Entity Type	Definition
61	party	A person or entity involved in a legal matter.
62	person	An individual or legal entity recognized by law as having rights and responsibilities.
63	petition	A formal written request submitted by an individual or group to a court, government agency, or other authority, asking for specific action.
64	petitioner	An individual or entity that initiates legal proceedings by filing a petition with the court.
65	plaintiff	The party who initiates a legal action by filing a complaint against another party, seeking compensation or specific relief.
66	police	Government officials tasked with enforcing laws, maintaining public order, and preventing crimes.
67	policy	A formal statement or plan by an organization, government, or authority that guides decision-making and actions.
68	position	A defined function or responsibility assigned to an individual within an organization, government, or legal context.
69	procedure	A systematic series of steps or formalities established by law or regulation for conducting affairs or business.
70	process	A structured sequence of steps followed by parties or courts to resolve disputes, enforce rights, or carry out legal actions.
71	property	A legal right to control, use, and transfer an item or resource, recognized by law.
72	prosecution	The legal process in which the government or state initiates proceedings against an individual or entity accused of violating the law.
73	provision	A specific clause, section, or stipulation within a contract, statute, or legal document that outlines terms, conditions, rights, or obligations.
74	publication	A written or printed work consisting of pages glued together within a cover, intended to contain a continuous body of text.
75	ratiodecidendi	The rationale or reason underlying a judicial decision, especially the principle or rationale that is binding.

S.No	Entity Type	Definition
76	regulation	A formally established rule or order issued by a governing authority to regulate conduct or activities within its jurisdiction, carrying the force of law.
77	respondent	A person legally required to reply to a notice or charge, typically the defendant in a trial.
78	role	A defined function or responsibility assigned to an individual within a legal context.
79	rule	A principle or directive established by a governing authority to regulate conduct or establish standards of behavior.
80	section	A subdivision within a statute or regulation, organized hierarchically to structure legal provisions.
81	sentence	A punishment or penalty imposed by a court on a person convicted of a crime.
82	statute	A written law enacted by a legislative body, having the force of law at the national, state, or local level.
83	suspect	A person regarded as having committed or being responsible for a criminal offense, but not yet proven guilty.
84	tax	A compulsory contribution imposed by the government on individuals or entities to generate public revenue.
85	victim	A person who has been wronged, injured, or otherwise harmed by the unlawful acts of another.
86	weapon	An object designed or used to cause physical harm or incapacitation to another person or entity.
87	witness	A person who has firsthand knowledge of facts or events and can provide testimony regarding them in a legal proceeding.

Chapter 7

Appendix-2

The expanded relations with their definitions are provided in tables from the next page.

S.No	Relations	Definition
1	acceptance	Acceptance refers to the act of agreeing to or receiving something, often in legal contexts such as contracts or settlements.
2	accompany	To attend or be present alongside another party in a particular matter or proceeding.
3	accuse	To formally charge or allege wrongdoing against another party in a legal proceeding.
4	accused	A person formally charged or alleged to have committed a crime or wrongful act.
5	accusing	Accusing refers to the act of making a formal allegation by one legal entity or person against another.
6	acquire	To obtain legal rights or title to something through lawful means.
7	acquitted	A defendant has been legally determined not to be guilty of the charges brought against them.
8	act	The performance of an action by one party influences or determines their legal obligations or rights concerning another party.
9	act_under	One legal entity acts or operates on behalf of or under the authority of another legal entity.
10	action	A formal proceeding in court to initiate legal relief or enforce rights.
11	acts_on_behalf_of	A party acts on behalf of another party within the scope of their authority or agency.
12	addressed	The legal concept X is addressed by legal concept Y when Y deals with X.
13	adjust	To modify or alter terms, conditions, or obligations under a legal agreement.
14	admit	To formally acknowledge or confess to the truth of a charge or claim in a legal context.
15	advanced	The relation 'advanced' indicates a specialized or higher level of knowledge, skill, or understanding within a particular legal domain.

S.No	Relations	Definition
16	affect	When one legal concept influences or impacts another in altering rights, duties, or responsibilities.
17	affiliate	A business relationship where one entity has a significant ownership or control over another, typically through shared financial interests.
18	affirmation	The formal recognition of the validity or enforceability of a legal document, decision, or declaration.
19	afford	To afford denotes having the financial means or capacity to pay for something, considering the individual's circumstances and without causing undue hardship.
20	against	One party opposes or acts in conflict with another party's rights or interests.
21	agency	A relationship where one party (agent) acts on behalf of another (principal), with the agent's actions legally binding the principal.
22	agent	An individual or entity acting on behalf of another party within a legal context.
23	agreement	A mutual understanding or arrangement between two or more parties to perform specific actions or refrain from certain behaviors, forming the basis of legal obligations.
24	allegation	An allegation involves the assertion by one party that another has committed a specific act.
25	allowance	The grant of permission or exemption from a restriction under a statute or regulation.
26	analysis	The detailed examination of legal information to derive meaning and draw conclusions.
27	answer	A formal written response by a defendant to a plaintiff's allegations in a legal case.
28	appeal	A request made to a higher court or tribunal to review and overturn a lower court's decision.
29	appeal	A formal request by one party to a higher judicial authority to review and overturn a lower court's decision.
30	arrest	The act of detaining an individual suspected of committing a crime by law enforcement authority.

S.No	Relations	Definition
31	assault	Assault occurs when one person intentionally causes another to experience a reasonable apprehension of imminent harmful or offensive contact.
32	assess	To evaluate or determine the legal validity, value, or implications of a legal concept by examining relevant factors or information.
33	assessment	The process of determining or evaluating a specific value, amount, or legal effect.
34	asset	A right or interest in property intended to provide a present or future benefit.
35	assign	To transfer ownership or rights from one party to another through a formal agreement.
36	assist	One legal concept provides support or aid to facilitate the implementation or enforcement of another.
37	associate	To form a connection or partnership with another entity or individual for mutual purposes or interests.
38	authority	Authority refers to the power or right of one legal entity or person to make decisions, enforce laws, or regulate actions within the framework of another.
39	authorityover	The power or right to direct or control the actions or decisions of another entity or individual.
40	authorize	To grant official permission or formal authority to someone to carry out specific actions or enforce laws.
41	authorize	To grant official permission or power to act under legal authority.
42	award	The amount of money granted to one party as compensation for loss or fulfillment of a contractual obligation.
43	based_on	The relationship where one legal concept or rule is founded upon or derived from another.
44	binding	An obligation or duty imposed by one legal concept on another.
45	breach	A failure to fulfill an obligation or duty under a contract or legal responsibility.

S.No	Relations	Definition
46	bring	To initiate or transfer the responsibility or authority for a legal matter or proceeding.
47	calculate	To determine a specific value or consequence by systematically evaluating relevant factors or applying legal principles.
48	call	A formal request by a judicial officer for an individual to appear in court.
49	cancel	To terminate or nullify an agreement or obligation.
50	cannot_claim	One party is prohibited from making a legal claim against the other.
51	carried	Refers to the transport of goods under an agreement where the carrier assumes responsibility for the items during transit.
52	carries	An estate or interest carries specific rights that may survive or transfer independently of the ownership of the underlying property.
53	case	A legal proceeding where a court addresses a specific matter between two or more parties.
54	case_related	Two legal cases that share common issues or parties.
55	caught	A legal concept describing the apprehension or detention of an individual by law enforcement.
56	cause	Establishing an event or action that produces a specific legal result.
57	certainty_of_fact	The degree of confidence required to establish a fact beyond doubt in a legal context.
58	challenge	A party challenges a decision when they contest its validity or correctness.
59	charge	A charge denotes the formal accusation brought by a prosecuting authority against a defendant.
60	citation	A reference or acknowledgment of a legal authority, case, or statute in written work to support an argument or conclusion.

S.No	Relations	Definition
61	claim	One party asserts a claim by stating its right or entitlement to something through a legal process or argument.
62	classification	The organization of legal concepts into categories based on shared characteristics or criteria.
63	close	Indicates the nearness or proximity between two legal concepts.
64	commission	A situation where one party acts on behalf of another to accomplish a specific task or transaction.
65	commit	When one legal concept commits another, it signifies that the former perpetrates or carries out the latter as a criminal act.
66	communication	The exchange of information or messages between two parties in a manner that creates mutual understanding and may establish legal rights or obligations.
67	complaint	A complaint refers to the act of formally accusing another party of wrongdoing to initiate legal proceedings.
68	compliance	The relation 'compliance' indicates that one legal concept is required to adhere to or conform with another.
69	conclusion	The final decision reached after analyzing the facts and law in a legal matter.
70	conduct	Conduct refers to the actions taken by a party that determine whether they are acting lawfully or unlawfully.
71	confer	To grant or impart a right, power, or privilege.
72	confine	To restrict or detain a person against their will under legal authority.
73	confirm	To validate or substantiate the accuracy and legality of a decision, claim, or fact.
74	conflict	The relation 'conflict' indicates an opposition or inconsistency between two legal concepts.
75	conveyance	The transfer of legal title or ownership rights in property from one party to another.

S.No	Relations	Definition
76	convicted	A person has been legally determined to be guilty of a crime.
77	conviction	The legal determination that an accused person is guilty of a crime.
78	corroboration	The process of supporting a fact or assertion with additional evidence or testimony.
79	court_proceedings	The sequence of formal steps and events carried out during a legal trial or hearing involving actions and rulings.
80	creation	The establishment of a new legal right, obligation, or entity.
81	crime	an unlawful act punishable by law.
82	custody	the legal right to care for and supervise a child, typically granted to parents under the terms of a court order.
83	date	The date specifies a particular time point in which an event or obligation occurs under the law.
84	dealing_with	The act of handling, addressing, or managing a particular legal matter or issue.
85	dealt_with	A legal authority or entity responsible for addressing or managing a particular matter.
86	death	The termination of one legal concept results from an event involving death.
87	decide	A judicial body formally determines the resolution of a matter, such as a verdict, sentence, or order.
88	decision	The resolution made by a court or tribunal to determine the outcome of a legal matter.
89	declare	To formally state or announce a legal right, obligation, or fact.
90	defence	A justification or excuse used by a defendant to reduce or eliminate liability for a wrongdoing.

S.No	Relations	Definition
91	defendant	The party accused of a legal wrongdoing or breach who is required to answer the plaintiff's claims in court.
92	define	One concept defines another when it provides a detailed explanation or specifies the scope and meaning of the latter.
93	delegate	The transfer of authority from one party to another to act on their behalf within specified limits.
94	deliver	To transfer or hand over something, such as property or documents, from one party to another as required by law.
95	demand	A formal request for specific action or compensation.
96	deny	One party refuses to accept or recognize the validity or enforceability of another's claims or rights.
97	depose	To compel a witness to provide sworn testimony in a legal proceeding.
98	deprive	To cause another person to lose a right, entitlement, or possession through wrongful conduct.
99	designation	The formal naming or appointment of a person to a specific role or position under the law.
100	determination	A formal decision made by a legal authority to settle a matter or issue.
101	dictate	One legal concept imposes or requires the existence or implementation of another.
102	did_not_commit	One party is not legally responsible for the actions or obligations of another.
103	direct	A direct relationship exists when one legal concept is immediately connected to another without intermediaries.
104	disallow	Disallow refers to the act where one legal concept restricts or prohibits another from being implemented or enforced.
105	discharge	To release or rid oneself of a legal obligation or liability.

S.No	Relations	Definition
106	disclose	To reveal information, required by law or contract, from one party to another.
107	discriminate	the unjust or unequal treatment of an individual based on protected characteristics such as race, gender, religion, or age.
108	dismiss	To dismiss a case means to terminate its proceedings without determining guilt or liability.
109	dispose_of_case	A judge or court settles and finalizes a legal case.
110	disregard	Disregard refers to ignoring a fact or circumstance because it does not qualify as binding or valid under the law.
111	distribution	The transfer of assets or rights from one party to another based on legal rules or agreements.
112	does_not_alter	The first concept does not serve to amend, modify, or change the second concept in any legal capacity.
113	does_not_apply_to	A rule or law has no effect on a particular person, entity, or situation.
114	does_not_have	Signifies the absence of possession, entitlement, or qualification between two legal concepts.
115	does_not_impose_obligation	The relation indicates that one party or entity does not create any mandatory duties or requirements for another.
116	does_not_prohibit	Indicates that one legal provision does not prohibit or restrict another.
117	does_not_require	One legal concept does not necessitate the existence or occurrence of another.
118	domicile	Domicile refers to a person's permanent home or principal establishment where they have the most significant connections.
119	duty	Duty refers to an obligation owed by one legal entity to another under the terms of a contract, statute, or agreement.
120	duty_imposed_by	A legal entity or authority imposes an obligation or responsibility (duty) upon another party.

S.No	Relations	Definition
121	effect	The consequence or outcome resulting from a legal rule, statute, or action.
122	effective_date	The date when a legal provision, agreement, or statute becomes legally binding and enforceable.
123	eject	The act of legally compelling an individual to vacate a property or premises.
124	eligibility	The condition that determines whether one legal concept grants access or entitlement to another.
125	enactment	The formal process by which a legislative body introduces and approves legislation to create or amend laws.
126	encroachment	The act of unlawfully entering or occupying someone else's land or property without authorization.
127	enforcement	The act of ensuring compliance with a rule or obligation.
128	engagement	the process by which two legal entities establish a formal agreement or obligation.
129	entitled_to	One legal concept grants another legal concept the right or privilege to receive something.
130	evidence	Evidence refers to information or material presented to support a claim, fact, or assertion in a legal context.
131	evidence_for	A serves as evidence supporting B.
132	examination	The process of thoroughly analyzing or inspecting something under legal scrutiny.
133	examination	An official process to assess or verify something, typically conducted by a court or tribunal.
134	exclude	The exclusion refers to circumstances where one legal concept is removed or not included within the scope of another legal concept.
135	exclusion	Exclusion refers to the act of removing or preventing someone or something from being included, recognized, or entitled to participate in a legal matter.

S.No	Relations	Definition
136	executed	the carrying out or putting into effect of a legal obligation, agreement, or document.
137	exercise	The active use or invocation of a right, privilege, or power under specified conditions.
138	express_opinion	One party formally states their view on another party's legal rights or obligations.
139	extend	A legal authority extends another concept, typically a deadline or statute.
140	facilitate	To provide support or means that make possible or easier the implementation or enforcement of a legal concept.
141	failure	The occurrence of an unforeseen event that prevents the fulfillment of a legal obligation.
142	falls_under	The first legal concept is included within or constitutes a part of the second legal concept.
143	false	A statement that is not true or does not correspond to reality.
144	fault	A failure to act as required by law, resulting in liability.
145	file	To create, store, or submit a formal record of legal actions.
146	find	To find means for a court or tribunal to reach a decision or determination on a particular issue or fact in legal proceedings.
147	form	The relation 'form' indicates how one legal concept is structured or established through another.
148	frame	The semantic meaning of the relation 'frame' is that it defines the boundaries or context within which two legal concepts are understood or applied.
149	hears	A direct personal knowledge of a fact, rather than information learned from another source.
150	heir	A person who legally inherits property or rights after another's death.

S.No	Relations	Definition
151	held_that	One party is legally responsible for another's debt or obligation.
152	impose_penalty	To apply or enforce a punishment or sanction.
153	imposes_restrictions_on	The relation indicates that one legal concept imposes limitations or constraints on another.
154	imposes_tax	A government authority mandates the collection of taxes from individuals or entities.
155	incorporate_law	The process by which a lower or subsidiary law becomes part of a higher or primary legal framework.
156	influence_legal	One legal concept affects or induces another legal concept to act or behave in a particular manner within the bounds of the law.
157	information_disclosure	The act of sharing specific information by one party to another as required by law or contractual obligations.
158	initiate_action	It refers to starting or beginning a legal process or proceeding.
159	injury_claim	A legal demand for compensation made by an injured party against another party alleged to have caused harm or damage.
160	institution_legal	The relationship where one legal concept is established or operates under the authority of another.
161	intent	Intent refers to the state of mind or purpose that a person has when performing an action, establishing their knowledge or desire regarding its consequences within legal contexts.
162	intent_knowledge	The mental state where a person knowingly acts with a specific purpose or intention, often influencing the legal consequences of their actions.
163	interaction	The relation captures how one legal concept influences and is influenced by another legal concept through mutual effects or interdependencies.
164	interest	Interest refers to a legal right or entitlement that one party has over another's property or asset.
165	interest_in	A legal claim or entitlement to use, possess, or otherwise benefit from a particular thing or right.

S.No	Relations	Definition
166	interpretation	The process of explaining or clarifying the meaning of a legal text.
167	intervention	A third party's action to influence a legal proceeding due to having an interest in the matter.
168	introduction	The act of formally presenting or bringing a legal concept, principle, or matter into a discussion or proceeding.
169	invalidation	Invalidation refers to the act of rendering another legal concept without legal effect.
170	investigation	A formal inquiry conducted by authorities to gather evidence regarding suspected illegal activities.
171	investment	The allocation of capital into an enterprise or project with the expectation of generating economic gain.
172	invitation	An invitation refers to a formal request by one party to another to engage in a specific action, which may establish the basis for a legal obligation or agreement.
173	involved_in	One party participates or engages in a legal proceeding or action.
174	involvement	The act of participating in or being associated with a wrongful act, thereby making the individual liable or responsible.
175	is	Describes a direct equality or identity between two legal concepts.
176	issued	A document is issued when it is created or given authority by an official entity.
177	issuing	The act of creating and formally authorizing a legal document or instrument to be officially produced and put into effect.
178	judge	A judge has the authority to preside over and make decisions regarding legal matters involving another individual or entity.
179	judgment	A formal decision made by a court or tribunal resolving the issues of a case.
180	judicial_authority	The power of a court to interpret and apply laws, make binding decisions, and enforce legal principles.

S.No	Relations	Definition
181	jurisdiction	Jurisdiction refers to the authority of a court to hear and decide cases within a specific geographic area or over particular types of legal matters.
182	kill	The act of intentionally causing another person's death through unlawful means.
183	knowledge	The understanding or awareness of facts necessary for establishing liability or intent in legal contexts.
184	liability	The legal responsibility of a party for harm or loss caused by their actions or omissions.
185	maintain	One legal concept ensures the continuation or enforcement of another.
186	member_of	Indicates that one entity is part of another group, organization, or body.
187	mention_in_document	One legal concept is cited or referenced in the context of another.
188	modify	The relation 'modify' indicates that one legal concept alters or changes another.
189	monetary_value	The amount of money assigned to assess the financial worth of a right or obligation under a legal agreement.
190	non_compliance	The failure to adhere to legal requirements or obligations.
191	notice	Notice is a formal or official communication of information, often required by law.
192	obligate	Signifies a binding legal duty imposed on one party by another through enforceable terms or law.
193	obligation	An obligation represents a duty or burden enforceable by law that requires one party to perform specific actions or refrain from certain behaviors.
194	observe	To take notice of or pay attention to a legal requirement, rule, or condition.
195	offer	An offer is a proposal made by one party to another that, if accepted, creates a binding agreement.

S.No	Relations	Definition
196	omission	Omission refers to the failure to act when there was a legal duty or obligation to do so.
197	on	One concept imposes a condition or requirement on another.
198	operate	To operate means one entity controls and manages another within established legal frameworks.
199	opinion	An opinion is a legal conclusion or assessment provided by a judge, jury, or legal expert that guides judicial decisions or provides authoritative advice.
200	oppose	One party resists or challenges another's position, rights, claims, or actions within a legal framework.
201	order	A directive issued by a court or judicial authority requiring specific actions or behaviors to be carried out, enforceable under the law.
202	originate_from	The legal relationship where a right or obligation arises from a specific source, such as a contract, statute, or precedent.
203	outside	The first legal concept is not included within or subject to the jurisdiction or application of the second.
204	overrule	Overrule occurs when a higher authority sets aside a prior decision due to an error in the interpretation or application of the law.
205	oversaw	To have authority or responsibility over another person's actions or decisions.
206	oversees	To monitor and ensure compliance with laws or policies.
207	owe	To have a debt or legal obligation to provide something of value.
208	own	To have lawful possession, authority, or control over a person, entity, asset, or right.
209	paid	An exchange of value provided in return for goods, services, or consideration under a legal agreement.
210	part	One entity serves as a component or portion of another entity.

S.No	Relations	Definition
211	participant	An individual or entity involved in a legal matter, holding rights and obligations under the law.
212	parties	The Parties clause identifies the individuals or entities involved in a legal matter.
213	party	A party is an individual or entity involved in a legal proceeding who has legal rights or obligations related to the matter.
214	payment	The transfer of money or property from one party to another in fulfillment of an obligation or agreement.
215	penalty	A penalty is a punitive measure imposed by law for the violation of legal rules or regulations.
216	perform	To carry out or execute an action, obligation, or task as required by law.
217	permit	A permit is a formal authorization or license granted to allow a specific action that might otherwise be restricted or prohibited.
218	petition	A formal request made to a court or authority.
219	place	The location where legal events occur or obligations are tied.
220	position	The relative status or arrangement of one legal concept concerning another.
221	possess	One party has rightful ownership or control over another entity as their property.
222	power	Power denotes the authority granted to an individual or entity to make decisions, enforce rules, or take actions within a specific scope.
223	prescribed	Indicates that a legal requirement or obligation is established by law.
224	present	One legal concept is currently in effect or existing relative to another.
225	procedure	The formal steps and processes established by law that must be followed in any legal action or proceeding.

S.No	Relations	Definition
226	produce	To generate, create, or bring into existence a legal document, obligation, or right under specific circumstances.
227	prohibit	To establish a legal restriction or constraint that makes an action unlawful.
228	property	Property refers to the bundle of rights in and to something tangible or intangible, including possession, use, control, exclusion, derivation, and disposition.
229	prosecution	The legal process initiated by the government or a designated authority to enforce laws against individuals or entities suspected of violations.
230	protect	To safeguard or uphold the rights or interests of a party against harm or interference.
231	provide	Provide refers to the act of furnishing, delivering, or making available something, such as information, goods, or services, from one party to another.
232	provision	A legal provision refers to the establishment or creation of rights, duties, powers, or restrictions within a legal document.
233	publish	To officially release legal information for public access.
234	purchase	The act where one party buys something from another in exchange for consideration.
235	pursuance	The continuation or furtherance of official duties or powers.
236	quash	To invalidate or nullify a legal order or decision, rendering it ineffective.
237	quote	To repeat verbatim from a legal authority or document while attributing the source.
238	raise_before_court	One party formally presents an issue, motion, or objection to the court for its consideration during a legal proceeding.
239	receive	One legal concept obtains or gains possession of something from another.
240	record	A record refers to the official written account of legal proceedings, decisions, or actions.

S.No	Relations	Definition
241	recover	To obtain something lost or awarded through a legal process.
242	reduce	One party diminishes or lessens the rights, obligations, or penalties of another.
243	refer	To make reference to or cite something.
244	refuse	The act of intentionally withholding consent or denying a request.
245	regard	Consideration or attention given to one legal concept in relation to another.
246	registered	The process by which a company officially records its existence and operations within a specific jurisdiction.
247	regulated	The term refers to the control or oversight exercised by one legal entity or authority over another, ensuring compliance with established laws, rules, or standards.
248	reject	Indicates that one legal concept refuses to accept or acknowledge another.
249	relate	To establish or describe the connection or association between two legal concepts.
250	release_on_bail	The defendant is released from custody under certain conditions as part of the bail process.
251	reliance	Reliance refers to the justified dependence by one party on the statements or conduct of another party in legal contexts.
252	remove	To legally transfer authority from one party to another.
253	rendered_judgment	The formal decision made by a court or tribunal in a legal proceeding, officially pronouncing the outcome.
254	repeal	The act of officially canceling or invalidating a law, statute, or regulation, thereby removing its legal authority and application.
255	replace	Replace indicates that one legal entity takes the place of another in a specific context.

S.No	Relations	Definition
256	report	A report relation indicates that one legal concept provides information or documentation about another legal concept.
257	represent	Represent means for one legal entity or person to act on behalf of another, conveying their interests or rights.
258	representation	A legal relationship where one entity acts on behalf of another.
259	reprimand	An official rebuke from a legal authority to an individual or entity for misconduct.
260	request	A formal or informal expression of a need or demand directed to another party in legal proceedings.
261	require	One party is legally compelled or obliged to perform specific actions or meet certain conditions imposed by another party.
262	response_obligation	A legal requirement for one party to formally respond to allegations or claims made by another party in a legal proceeding.
263	restriction	The application of a limitation or condition that controls or limits how one legal concept operates in relation to another.
264	result	The Result relation indicates the consequence or outcome produced by one legal concept concerning another.
265	review	A formal examination and evaluation of a decision or action by a higher authority, potentially leading to modification or overturning.
266	right	A justifiable claim or entitlement that is enforceable by law.
267	role	The relation 'role' defines the specific responsibilities, rights, and duties that each entity is assigned within a legal context.
268	rule	A rule establishes standards of conduct or procedures that govern behavior in legal contexts.
269	sanction	A formal penalty imposed by a court or authority to enforce compliance with a law, rule, or obligation.
270	save	To preserve or protect from harm, loss, or penalty.

S.No	Relations	Definition
271	scrutiny	A formal and detailed examination of actions or decisions to assess their legality or appropriateness.
272	search	The examination or exploration of a person's property or communications to locate evidence.
273	section	A section refers to a subdivision within a larger legal document, such as an act or regulation.
274	secure	To obtain or guarantee rights or protections.
275	seek	To seek means to actively look for or attempt to obtain something, often rights, remedies, or compensation.
276	sentence	A sentence relates an accused individual to the specific punishment they are assigned.
277	serve	One legal concept provides the means or mechanism necessary for another legal concept to be fulfilled or achieved.
278	shall	In legal contexts, 'shall' is used to create mandatory obligations or duties that must be fulfilled.
279	show	The first concept serves as evidence to prove the second concept.
280	sign	One legal concept is authenticated or agreed to by another.
281	similar	Indicates that two legal concepts are alike or share comparable characteristics.
282	slander	A false and defamatory statement made orally or in a transitory form that harms another's reputation.
283	state	Describes the quality or condition of a thing in law.
284	status	The current legal state or condition of a subject.
285	strike_action	A collective work stoppage by employees to protest or negotiate terms with an employer.

S.No	Relations	Definition
286	subject	A person or entity that is the focus of a legal proceeding or obligation.
287	submit	Indicates the formal handing over of a legal document or request to an authority.
288	succession_rights	The legal entitlements and interests that pass from one person to another upon death.
289	sue	The act of initiating legal proceedings against another party to seek compensation or remedy for perceived wrongs.
290	supersede	To take precedence over and replace a prior legal concept.
291	supervise	One legal concept has the authority to monitor, direct, or regulate the actions of another to ensure compliance with established rules or standards.
292	supplementary	Indicates that one provision or document is added to or supports another, but does not replace it.
293	support	The relation 'support' indicates that one legal concept furnishes the basis or justification for another.
294	suspect	A person identified by law enforcement as having reasonable grounds to believe may have committed a crime.
295	suspend	The temporary cessation of a right, privilege, or obligation as a penalty or measure until specified conditions are met or circumstances change.
296	tax	A tax is a compulsory contribution imposed by a government authority to fund public services or governmental functions.
297	terminate	To end legal obligations between parties with no further responsibilities.
298	testify	The act of providing evidence or a statement under oath in legal proceedings.
299	testimony	The information provided by a witness under oath during legal proceedings.
300	transfer	The act of conveying rights or obligations from one party to another.

S.No	Relations	Definition
301	transfer	To convey or move ownership, title, or rights from one party to another.
302	treat	A treats B when A takes action or omits to act in a manner that affects B's rights, obligations, or interests under certain conditions.
303	trial	A formal proceeding where evidence is presented to determine the outcome of a legal case.
304	urge	Compel or press for immediate legal action regarding a matter.
305	use	The application or employment of one legal concept in relation to another, such as using intellectual rights over tangible property.
306	validate	To confirm that something is legally valid or enforceable.
307	validation	The process of confirming that a legal document, action, or decision complies with applicable laws and standards.
308	value	Value refers to the worth or benefit of something in terms of money, goods, or services.
309	versus	Indicates opposition or conflict between two legal concepts or parties.
310	vested	Refers to a legal right or interest that has become fixed and cannot be revoked.
311	victim	A person who has suffered harm, damage, or loss as a direct result of the unlawful actions or omissions of another party.
312	violate	Intentionally disregard or break a law, contract, or other legal requirement.
313	visit	The act of one party or entity lawfully entering or being present at another party's location for a specific purpose.
314	was_allowed	The permission granted to an individual or entity to perform a specific action under certain conditions.
315	was_charged	to have formally accused someone of committing a specific offense

S.No	Relations	Definition
316	was_conducted_by	Indicates that one entity carried out or managed a particular process or action.
317	was_in	The individual was present at a specific location during relevant events or at a specified time.
318	was_involved_in	Indicates participation in a case, matter, or transaction.
319	was_made_by	Indicates that one legal entity or instrument was created or produced by another party.
320	witness_relation	The connection between a witness and another party or element in a legal proceeding.